

SAE Baja Capstone Final Design Report 2024

Prepared for: Emily Artale and Lauren Cooper



**PARTNERSHIP PROGRAM
COMPUTER SCIENCE & ENGINEERING**

Statement of Disclaimer

This paper discusses the 2023-2024 Western Colorado University, Colorado University - Boulder Partnership Capstone Baja SAE final design. The paper starts with introducing the project, followed by key design constraints. After the design constraints, the team discusses the final design for the Baja car broken down by subsystems. The paper is finished with a timeline of when the team finished tasks related to building the car and the competition in Gorman, CA. Note that the design is finalized. Since this project is a result of a class assignment, it has been graded and accepted as a fulfillment of the course requirements. Acceptance does not imply technical accuracy or reliability. Any use of information in this report is done at the user's risk. These risks may include catastrophic failure of the device or infringement of patent or copyright laws. Western Colorado University, University of Colorado at Boulder, and its students, faculty, and staff cannot be held liable for any use or misuse of the project.

Abstract

This is the final report and summary of the WCU/CU Boulder 2023-2024 SAE Baja Capstone project. The report's experiences and findings reflect real work fabrication in combination with FEA analysis. The final section is a wrap-up with experiences that the team has learned from going to the SAE Baja California competition. The report follows the design process from ideation and constraints to prototyping, moving into fabrication, and finishing out with testing. We will then cover what we learned from attending the SAE competition in California and our takeaways.

Table of Contents

<i>Statement of Disclaimer</i>	<i>2</i>
<i>Abstract</i>	<i>3</i>
<i>Table of Contents</i>	<i>4</i>
<i>Table of Figures.....</i>	<i>6</i>
<i>Table of Tables</i>	<i>7</i>
<i>Introduction</i>	<i>8</i>
<i>Background.....</i>	<i>8</i>
Goals	9
Outcomes	9
<i>Design Considerations</i>	<i>11</i>
Objectives On Hold	11
Stakeholder Feedback.....	12
<i>Design Constraints</i>	<i>13</i>
<i>Frame/Chassis.....</i>	<i>15</i>
Concept Development	16
Concept Designs	18
Concept Justification.....	19
Project Management	21
<i>Front Suspension</i>	<i>21</i>
Concept Development	21
Concept Designs	23
Concept Justification.....	24
Project Management	25
<i>Rear Suspension</i>	<i>25</i>
Concept Development	25
Concept Designs	26
Concept Justification.....	29
Project Management	29
<i>Drivetrain.....</i>	<i>30</i>

Concept Development	30
Concept Designs	33
Concept Justification.....	33
Project Management	33
<i>Steering.....</i>	<i>34</i>
Concept Development	34
Concept Designs	35
Concept Justification.....	37
Project Management	37
<i>Project Tasks</i>	<i>37</i>
Task 1. Finalize CAD Model	37
Task 2. Finalize Frame Fabrication.....	38
Task 3. Finalize Suspension Geometry and Fabrication.....	39
Task 4. Design/Fabricate Backing Plates.....	39
Task 5. Preliminary Testing of Vehicle	40
Task 6. Redesign and Rebuild Based on Testing.....	41
Task 7. Train Driver	41
Task 8. Bring the Vehicle to Race in Gorman, CA	41
<i>Project Deliverables.....</i>	<i>42</i>
<i>Timeline</i>	<i>43</i>
<i>Competition Takeaways</i>	<i>44</i>
<i>Competition Agenda</i>	<i>44</i>
<i>Technical Inspections.....</i>	<i>44</i>
General Tech Inspection.....	45
4x4 check.....	45
Dynamic brake check	45
Engine check.....	45
<i>Design Briefing Review Presentation</i>	<i>45</i>
<i>Packing List</i>	<i>46</i>
<i>Competition Culture</i>	<i>47</i>
<i>Communication</i>	<i>47</i>

General Takeaways.....	47
Qualifications.....	48
Appendix A – Results from Stakeholder Mapping	50
Appendix B – Results from Stakeholder Interviews	51
B.1 - Jayden’s Interview with Zak Trafton.....	51
B.2 - Teag’s Interview with Lauren Cooper	51
B.3 - Tyler’s Interview with Eric Ortolano	51
B.4 - Shelby’s Interview with Chandra Proffitt.....	52
B.5 - Deagen’s Interview with Greg Vanderbeek	52
B.6 - Gable’s Interview with Kaiya Firor.....	53
B.7 - Spencer’s Interview with Charlie Schayer	53
Appendix C – Results from Whole Systems Mapping.....	55
Appendix D – Design Evaluation Score Sheet Summary.....	56

Table of Figures

Figure 1: Chassis progress after the first year.....	8
Figure 2: Excitement Attributes vs Threshold Attributes.....	11
Figure 3: CAD model of frame and chassis.....	15
Figure 4: Final chassis with most components iso view.....	15
Figure 5: Final chassis with components front view	16
Figure 6: Current nose with blue tape showing the proposed cutaway.....	18
Figure 7: Cutaway of internal sleeve showing rosette welds.....	19
Figure 8: Visual explanation of camber and caster.....	22
Figure 9: King Pin Inclination.....	22
Figure 10: Scrub Radius Definition.....	22
Figure 11: Current front suspension model in the master assembly.....	23
Figure 12: Updated front suspension and master assembly.....	23
Figure 13: Upper Control Arm (UCA).....	24
Figure 14: Lower Control Arm (LCA).....	24
Figure 15: Orthogonal view of trailing arm.....	26
Figure 16: Front view of trailing arm.....	26
Figure 17: Top view of trailing arm.....	26
Figure 18: Trailing arm with hub and mounting tabs.....	27
Figure 19: Rear hub carrier.....	27
Figure 20: Rear hub carrier FEA.....	28
Figure 21: Rear trailing arm FEA.....	28
Figure 22: Rear view of rear suspension.....	29
Figure 23: Engine and transmission mounts.....	30
Figure 24: Isometric view of differential and suspension components.....	31
Figure 25: Front view of differential showing axle output angles and suspension angles.....	31

Figure 26: Bottom view of differential showing increased toe box area.....	32
Figure 27: Initial driveshaft location plan.....	32
Figure 28: Final driveshaft location.....	33
Figure 29: Steering assembly in SolidWorks (minus u-joints).....	35
Figure 30: The angles and lengths involving Ackerman steering	36
Figure 31: Implemented rack and pinion.....	36
Figure 32: Tech inspection passing stickers.....	44
Figure 33: 4x4 check.....	45
Figure 34: Dynamic brake test.....	45

Table of Tables

Table 1: SAE International Deliverable Timeline.....	10
Table 2: Design Constraints.....	13
Table 3: Project Timeline 1.....	43

Introduction

The Baja Society of Automotive Engineers (SAE) Capstone Project focuses on the ground-up design and manufacturing of an off-road vehicle. Our design will use inspiration from other teams nationwide, suggestions from our stakeholders, guidance from last year's Capstone group, and a rigorous rule book given to us by the SAE organization. Using SolidWorks, we will design all sub-assemblies for the vehicle and compile them into a fully defined CAD model. We will manufacture the vehicle using in-house machinery tools, and we will purchase components from outside suppliers. Once the Baja vehicle is operating, we will take the vehicle to Hartman Rocks for suspension and chassis testing. Basic vehicle maneuverability testing, such as steering and functionality of the drivetrain, will be conducted outside the Rady building. The data from testing will validate any changes that may need to be made. Thus, in the redesign stage, we will find problematic areas of the vehicle and make improvements where necessary to create a safe and simple design.

Background

SAE International is a worldwide organization dedicated to serving as one of the top centers of knowledge in the engineering field. With the collaboration of over 145,000 engineers and technical experts, SAE fosters the spread of knowledge and expertise across various industries.

SAE has several competitions throughout the year, including the Baja competition. According to the official SAE Baja website, "[Baja SAE](#) challenges engineering students to design and build an off-road vehicle that will survive the severe punishment of rough terrain and, in some competitions, water. As in real work situations, these future engineers work together as a team to discover and resolve technical challenges in design, testing, manufacturing, and business issues. The most rugged of all the competitions, Baja SAE also gives students the first-hand challenge of pursuing their passion while managing real-life demands and priorities.



Figure 1: Chassis progress after the first year

Each team aims to design and build a prototype of an all-weather, rugged, single-seat, off-road recreational vehicle intended for sale to the nonprofessional weekend off-road enthusiast" [1].

Capstone members must function as a team to not only design, build, test, promote, and race a vehicle within the limits of the rules, but also to generate financial support for the project and manage their educational priorities.

As stated in the rule book in Appendix D, Article 2, Section B.2, every competitor must use a [14hp Kohler Command Pro CH440](#) engine with a modified restrictor plate to compete.

Restricting the engine encourages teams to focus on designing mechanical interfaces between components instead of making engine modifications.

This year's SAE Baja Senior Capstone project continues last year's Capstone project, where the team designed and partially completed the chassis manufacturing. The previous team designed portal axles and manufactured one to completion, but three more must be machined and assembled. Last year's team completed some CAD models, but as a team, we will need to assemble and finalize a CAD model before manufacturing the rest of our car.

Goals

Note: The following goals were what our year of capstone strived to achieve, the following year's goals may be similar or completely different but should not aim to replicate these exactly:

To understand the goals of the Baja Capstone leading into this previous year, we engaged seven stakeholders, including students, staff, sponsors, and industry professionals. The interviews taught us what the staff and community wanted to see from the BAJA Capstone project.

Common feedback we heard included:

- Finalize CAD model assembly.
- Design an adaptive suspension system.
- Manufacture in-house.
- Manufacture portal axles or switch to a more straightforward design.
- Test rigorously.
- Understand the rulebook.
- Allocation of work on the project.
- Prioritize safety.

Our main goal was to get the car on the ground and drivable. We shared this with all stakeholders. However, we had another important goal: we need to ensure that next year's team is set up to succeed.

Outcomes

Initial: This project will have multiple outcomes as we have deadlines and deliverables for the Capstone class and separate deliverables for the SAE organization. We plan to stay ahead of these dates and manage our time effectively. Class deliverables should not conflict with SAE's deadlines for the rest of this semester. Next semester, we will hold separate conversations with Capstone professors to work around these.

After the second semester, our overarching outcome of both semesters continued to be the driving vehicle and a complete CAD model with every component in it. This remained true and was not affected by changing capstone deliverables or SAE deliverables; however, throughout the semester, multiple conversations were had with professors and advisors, which changed the course of our capstone deliverables to fit the schedule and workload of the SAE deliverables.

Lauren has now planned on changing the way Baja's capstone deliverables look to better suit Baja's high demand from SAE while also meeting expectations for CU Capstone. Reach out to Professor Lauren Cooper regarding specifics with what is expected of Capstone deliverables for following years.

Table 1: SAE International Deliverable Timeline

Kohler Engine Orders Open	Friday, December 1, 2023
Kohler Engine Orders Close	Saturday, April 1, 2023
Business Event RPF Submission Due	Wednesday, December 20, 2023
Roll Cage Document Packet Due	Thursday, February 1, 2024
Written Cost Reduction Report Due	Monday, March 4, 2024
Business Present Form Due	Monday, March 11, 2024
Online Cost Prototype Due	Monday, March 18, 2024
Competition Design Report Due	Monday, March 25, 2024
Virtual Business Presentation	Thursday, March 28, 2024
Fast Track Roster Affiliation Due	Tuesday, April 9, 2024

Design Considerations

Kano Model

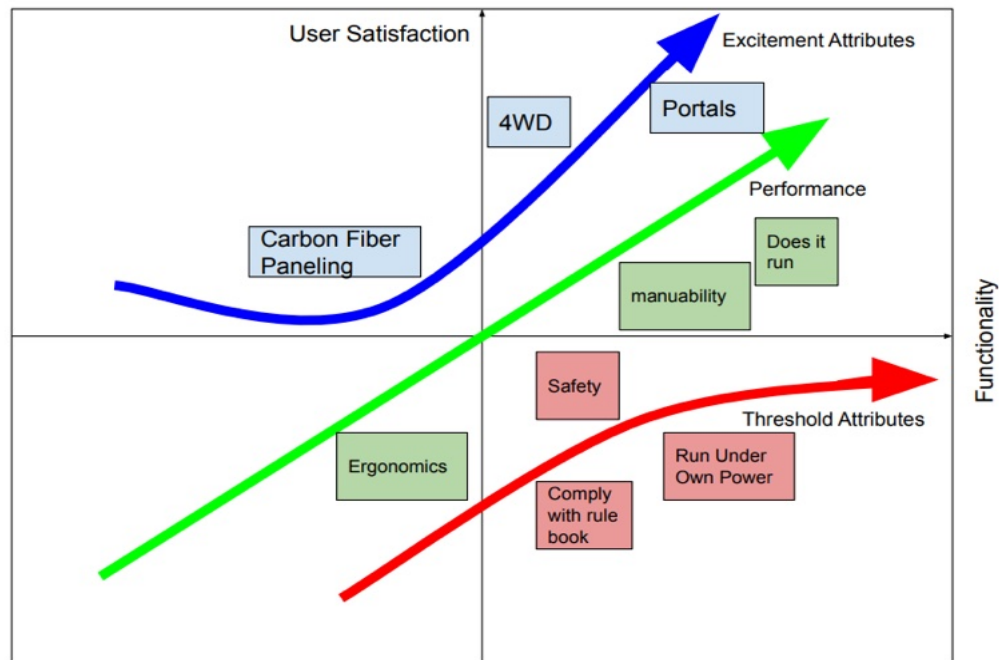


Figure 2: Excitement Attributes vs Threshold Attributes

The Kano Model illustrates a hierarchy of priorities in our design. It can be seen in Figure 2 that, at a minimum, our top priority is complying with the rulebook attached in Appendix D. At the competition, every car must go through a technical inspection. If the car does not pass technical inspection, it cannot race until changes are made. As a Capstone group, we have made passing this inspection our most crucial restriction in the design stage. Every component will have been reviewed with the newest version of the SAE Baja Rule Book.

Note: 4WD was later required by SAE and is therefore no longer an excitement attribute but is now required to pass tech inspections and is, therefore, a threshold attribute.

Objectives On Hold

The Baja Capstone team has decided to forgo some extra features that are not necessary to get the car race ready in April. These features would be great additions in future years when the team has the time to undertake the projects.

Due to the complexity of portals, the Baja Capstone team has decided to design the car around no portals this year. The portals would require 80-plus hours of machining time. This estimate is a large amount of time that can be used more effectively to produce a running and driving car. Another reason the team put the portals on hold was due to testing equipment. The team has not been able to devise a testing method that simulates the cyclic and impact loading that the Baja

car is expected to experience under normal operation. The current team felt that the portals are better suited for a class that can dedicate their entire year to building and testing the portals, as well as modifying the car to accept the portals. This process would be after the car has been run at multiple races successfully and survived the endurance race.

Stakeholder Feedback

The SAE Baja Capstone project takes feedback from stakeholders very seriously as it is not only an engineering project but also has a significant impact on the school and community. Western Colorado University and community members have generously contributed to this project, mainly through donations from past fundraisers. As we value our contributors, we want to consider feedback from our key stakeholders. We conducted interviews (Appendix B), and the common feedback we received can be summarized with the following key points:

- Utilizing off-the-shelf products to support car production.
- Recycling car components when possible.
- All parts must be easily removable in order to be serviced in the field.
- Maintain good communication with our team and Lauren about events, manufacturing, deliverables, and deadlines.
- Plan for a task to take 50% more time than you expect.
- Portal axles.
 - At least 80 more hours to manufacture three more portal axles
 - We can test portal axle life with a motor but not under load or impact

The Baja Team has decided to take a few key points into consideration as we move forward. These points include:

- Utilizing off-the-shelf products to support car production.
- All parts must be easily removable in order to be serviced in the field.
- Maintain good communication with our team and Lauren about events, manufacturing, deliverables, and deadlines.
- Plan for a task to take 50% more time than you expect.

The team has decided on the above design considerations as they were topics brought up multiple times. The team also feels that these aforementioned design constraints can make or break the project, making them mission-critical.

After completing this project there are some key points from stakeholder interviews that we believe should be emphasized more than others:

- Plan for a task to take 50% more time than you expect, this could be taken to the next level, and assumed that projects will take double the time you expect. From design, prototyping, manufacturing, and then solving problems in the final part, you can expect that to take up to 2x the time you expected.
- Maintain good communication with our team and Lauren about events and manufacturing, deliverables, and deadlines. This is a large project with many aspects to it, keeping good communication with the group and especially Lauren will allow this

project to go more smoothly and eliminate the need to problem-solve deadlines before it's too late.

We included a whole system map to gain a big-picture view of the project (Appendix C). This map included important vehicle subsystems that were traced back to their origin. Key design considerations learned from **whole systems mapping** include:

- Plan for increased supply chain lead times.
- Take into account how future teams will reuse and incorporate parts we use this year.
- Involve club members to help manufacture the car.
- Layout each subsystem in an easy-to-read format.
- Understand everything that needs to be placed into the car for use to reach our goal.

Additional design considerations may be defined during product development. For more information, see *Appendix A* and *Appendix C*.

Design Constraints

The BAJA SAE Rule Book has outlined various design constraints to which the BAJA car must conform. Table 2 outlines the design constraints for all of the car's subsystems. *NOTE: The table below is a summary of the official rule book. Each Design Constraint is hyperlinked to the section it refers to. The full version B of the rule book can be found in Appendix D - Collegiate Design Series Baja SAE Rules 2024 Revision B, with the most up-to-date rule book linked in Appendix E.*

Table 2: Design Constraints

Design Constraint	Description
<u>General Design Requirements</u>	The BAJA car shall be designed so that any driver whose height and weight fall within the lower 5% of women and the top 95% of men can operate the car. The car shall be designed to run on rough land, rocks, and mud. The car shall be able to work in all-weather situations. The car shall have four wheels that can be powered (4WD/AWD). The car shall fit on a track designed for a car measuring a maximum of 64 inches wide and 108 inches long.
<u>Engine</u>	The car shall use an unmodified Kohler CH440 with a BAJA SAE restrictor plate. The engine shall be governed to 3,800 RPM.
<u>Roll Cage</u>	The roll cage and frame shall be built to protect the driver during regular operation and in the event of a crash. The roll cage shall be triangulated in design with gussets for additional strength.
<u>Driver Restraint</u>	The driver shall be restrained with a five-point harness system. The restraint system shall be SFI or FIA-certified. The restraint system must

	have a tag dated within three years of January 1 of the competition year or have a “Valid Until” date later than the competition.
Driver Equipment	The driver shall wear a SNELL-approved one-piece motocross helmet with a chin guard. The driver shall wear motocross-style goggles with a full elastic band. The driver shall use SFI-certified neck support.
Fuel System	The entirety of the fuel system shall be within the roll-cage envelope so as to be protected by the roll-cage in the case of an accident. A Pyrotec SFC1000 shall supply the fuel system. The car shall use the fuel tank cap provided with the Pyrotec SFC1000 or a Briggs and Stratton B4325GS.
Vehicle Controls	The car shall be equipped with hydraulic brakes on all four wheels. The brakes shall be applied with a single-foot pedal. The brakes must use two independent hydraulic circuits. The car's throttle shall be able to adjust the engine's throttle from 0% to 100% without electrical means.
Cockpit	The cockpit shall be enclosed to prevent the driver from coming in contact with any moving parts. The cockpit shall have a fire extinguisher installed using an approved mounting bracket.
Powertrain Guards	All power train components shall be shielded to prevent injury to the driver, track workers, and bystanders. All guards shall be made from a thickness of 0.06 inch (1.5mm) of steel or 0.12 inch (3.00mm) of aluminum.
Electrical System	The car shall have at least two approved engine kill switches. The car shall have an approved brake light and brake light switch. The brake lights shall be operational regardless of the engine kill switch position.
Tow Points	The car shall be equipped with a front and rear tow point. The tow points shall be strong enough to act as vertical lift points. The front tow point shall be able to pass a 2inx2inx8in block behind it. The rear tow point shall be a clevis hook matching the given dimensions.
Fasteners and Attachments	Fasteners shall be a minimum of SAE grade 5 or Metric grade 8.8. Two threads shall protrude from the fastener when installed. The fasteners shall be held captive by approved means.
Vehicle Identification and Markings	The car shall show its number as assigned by BAJA SAE. The car shall have its number in 3 locations on the car using an approved font and high contrasting color. The car shall have an approved transponder mounted to BAJA specifications.

Frame/Chassis

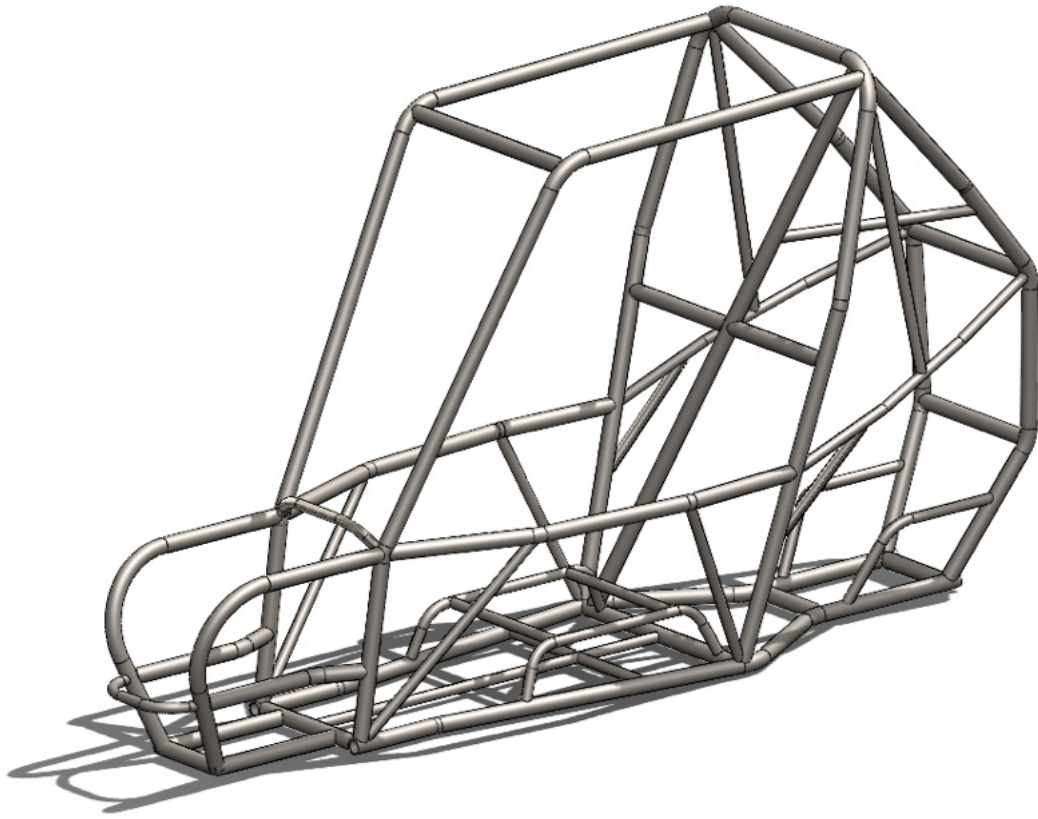


Figure 3: CAD Model of Frame and Chassis



Figure 4: Final chassis with most components iso view

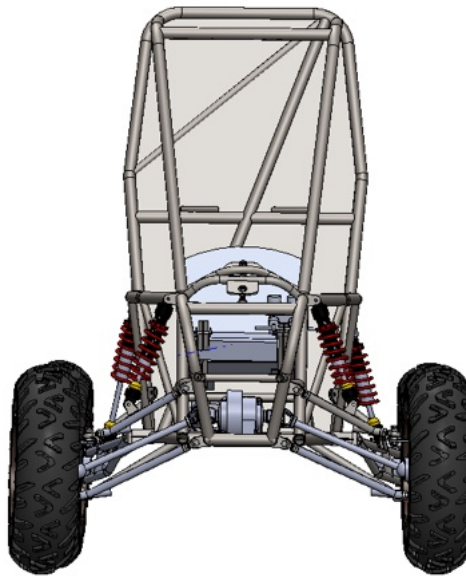


Figure 5: Final chassis with components front view

Concept Development

The Baja Capstone group has inherited a frame from a previous Capstone group. The following section is from the 2022-2023 Baja Capstone Group:

For a Baja vehicle, there are two main chassis frameworks. Rear braced and front braced, which define where the roll cage is braced. Many teams choose one framework to drive the bulk of their design but include elements of each framework in their design. After doing our research, we concluded that the front-braced or rear-braced vehicles are both structurally sound, and there was not a clear choice of one being better.

Since we are starting from the ground up, our design goals are different from most teams in the competition. The existing teams are looking to improve on the previous team designs while we are looking to create a baseline vehicle. Because of this, our goals for the chassis are shifted towards creating a functioning vehicle over prioritizing weight reduction. Another consequence of being a first-time team is that during the initial design process, there were a lot of unknowns about what parts and systems we would be using. After discussing these problems amongst the team and taking input from team members' personal experiences with racing vehicles, we decided on a primarily rear-braced framework. The driving factor for this decision was because of the wide range of options this type of bracing offered for mounting the engine and transmission. Requirements in Article 3 of the rule book drove the basic dimensions for the chassis.

These requirements include mandatory primary and secondary members, and a list of specifications for each required member. We are also using the required

driver space, engine and transmission space, and seat dimensions to inform our design. The driver template given in the Baja rules drives the dimensions of the roll hoop and driver compartment. The function of these dimensions is to ensure that the vehicle can fit a 95th-percentile male and a 5th-percentile female, as well as to assure the safety of the driver in the event of an accident. Considerations for the overall comfort of the driver also helped to determine the dimensions for the length of the chassis and the width of the driver compartment. Once we acquired the transmission from a Polaris Sportsman 500 and the Kohler engine, we used the dimensions of these components to design the rear of the vehicle, meaning everything behind the firewall.

The length of the transmission and clearance between the driven pulley and the firewall of the driver compartment determined the length of the rear compartment. Per rule B.6.3, the engine must be contained within the vehicle. This requirement and our decision to mount the engine on top of the transmission drove the dimensions for the height and width of the rear bracing. Considerations for the ease of insertion/removal of the engine and transmission after the frame has been built drove the width between members of the rear bracing. We added additional bracing to accommodate pre-existing mounting points on the transmission, but more will be needed for structural stability and to accommodate mounting for other components, such as the fuel tank.

The space required for pedals, steering components, wiring, and comfort for the driver's feet and legs determined the dimensions for the front foot box. While the shape for this section of the frame has been determined, exact dimensions cannot be assigned until these components are finalized. Many of these components have been researched but need to be fully modeled or integrated into the CAD model to ensure that there is no interference. Currently, we can foresee that the length of the box may need to be changed to accommodate the pedals, master cylinder, and steering components that need to be contained within this portion of the vehicle, as well as leaving room for the driver's feet.[2]

The current Baja Capstone group has reviewed the frame and ensured it meets current SAE Baja rules. The team has determined that the chassis is of a suitable quality to continue working with.

After completing the project: Looking back on the chassis for this project, there are many changes we would suggest. One major problem we encountered in our design this year was figuring out how to mount the suspension in the most optimal way. This challenge arose from the fact that our frame was already completed and not designed around the front and rear suspension. We recommend in future years designing the suspension first, and then designing a frame that meets the needs for mounting points and size around your suspension. Another recommendation we have for future years is frame materials. We were handed down a frame from the previous year that was made from tubing that greatly exceeded the requirements given by SAE. This means our frame was larger and weighed significantly more than other competitors' vehicles. This size and weight make it less efficient given the required engine that everyone must use.

Another follow-up to the design of the chassis is our last recommendation for a new frame design, and that is to pay very close attention to the SAE rulebook when designing your frame. We had to add multiple support members to meet the requirements of the rules and modify what we had. If closer attention had been paid to the rules book early in the design stage, we could have avoided this extra time, materials, and weight added to the car. Simplicity is not a bad thing. Straight members reduce the amount of time it takes to fit up the frame and reduce the amount of secondary members that are required to support the bends.

Concept Designs

Although the team has inherited a semi-welded chassis, a few changes were required. The most notable issue was in the nose of the car. The nose of the car must be able to hold a differential for 4-wheel-drive, a rack-and-pinion system for steering, and have some room left for pedals. The current frame does not have enough room to accommodate all the necessary components. In collaboration with the front suspension group and the steering group, the Baja team has decided to extend the nose of the car 3 inches. Through the extension of the car's nose, the steering team designed the steering for the car using an Ackerman steering geometry, the steering style of choice. This style of steering uses an arrangement of linkages to steer the car where each wheel needs to turn on a different radius to reduce pushing of the tire, increase grip, and extend tire life.

In order to extend the nose of the car, the team decided to cut the nose at the blue tape shown in Figure 4. The team rebuilt the nose with longer straight members to allow the 3-inch increase in length. The new nose was installed using an internal sleeve. The internal sleeve was a tight fit to the parent material, being welded together at the butt joint and at two rosette welds on both parent tubes, as shown in Figure 7.



Figure 6: Current nose with blue tape showing the proposed cutaway

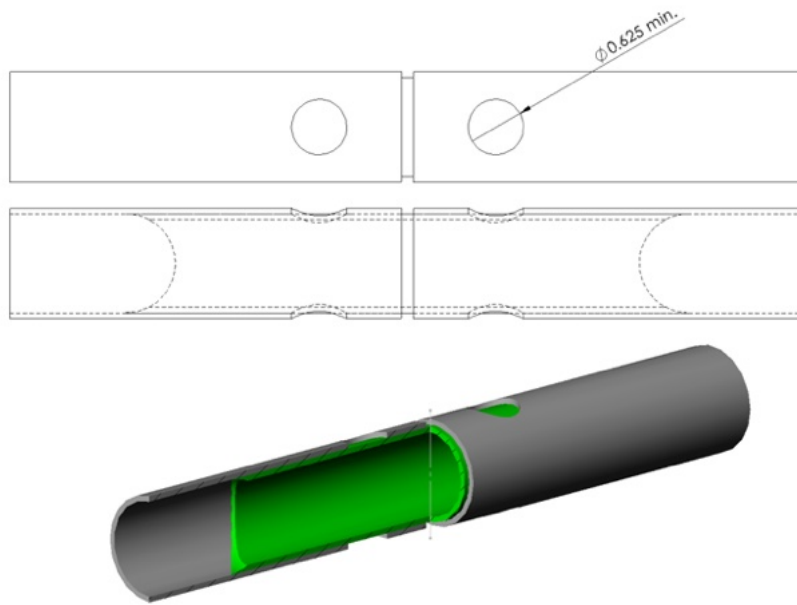


Figure 7: Cutaway of internal sleeve showing rosette welds.[1]

Concept Justification

The designs laid out above are approved under Collegiate Design Series Baja SAE Rules 2024 Revision B:

B.1.2 – Ergonomic Design: The roll cage is designed to accommodate all drivers who fall within the 95th percentile of men to the 5th percentile of women in the United States of America.

B.1.5.2 – Four-Wheel Drive / All-Wheel Drive: the chassis will be modified to accommodate a front differential.

B.3.2 – Roll Cage Structure: The car will be fabricated using tubular steel. The primary members will be a 1.5-inch outer diameter and 0.125-inch wall thickness steel drawn over mandril mechanical tubing (DOM tubing). The DOM Tubing will be bent in such a way as to maintain structural integrity, having no visible wrinkles, tears, or kinks.

B.3.2.1 – Member Requirement: The car will have no straight member whose length exceeds 40 inches between named points. Bent members will not have bends exceeding 30 degrees.

B.3.2.2 – Primary Member: The primary members will always start and end at named points.

B.3.2.2 – Secondary Members and Tubes: The secondary members will be made out of 1-inch-diameter, 0.035-inch-wall DOM.

B.3.2.4 – Additional Support Tubes: When primary members are longer than allowed, additional support tubes will be added. The support tubes will contact the primary members within 5 inches of their mid-point and terminate at a named point.

B.3.2.5 – LC – Lateral Cross Member: The lateral cross members will be straight tubes greater than 8 inches long. The lateral cross members will be made from the same 1.5-inch diameter—0.125-inch-wall DOM Tubing as the primary members.

B.3.2.6 – RRH – Roll Hoop Members: The roll hoop will be 29 inches or wider, measured 27 inches above the bottom of the seat. The RRH will be inclined less than 29 degrees off of the vertical.

B.3.2.7 – LDB – Lateral Diagonal Bracing: The LDB will be placed across the RRH, tying into the RRH within 5 inches of points A and B.

B.3.2.8 – RHO – Roll Hoop Overhead Members: The topmost members on the roll cage will be built to be in plane with each other. The member will be straight from the RRH to the LC at point C.

B.3.2.8.1 – Gussets for Lateral Clearance: The gussets will be made of 0.07-inch plate folded over to be greater than 0.125-inch overall thickness. The plates will be fully welded along the faces to be considered plates.

B.3.2.9 – LFS - Lower Frame Side Members: The LFS will start at the RRH and continue forward to point F, with another set of tubes starting at point F and moving forward to point E. The LFS will be made from primary tube material.

B.3.2.10 – SIM – Side Impact Member: The SIMs will start at point S and move forward to the front of the roll cage. The nose of the car shall have primary tubes running at the same height as the length of the nose.

B.3.2.11 – UTS -Under Seat Tube: There will be a tube made from the secondary member tubing to keep the driver in the car if the seat were to break away.

B.3.2.12 – FMB -Front Bracing Members: The FMB will be a solid tube with the RHO bent in front of point C, finishing at point F.

B.3.2.13 – FAB – Fore – Aft Bracing: The RRH will be restrained from rotation and bending at points B and C with the gussets used for lateral clearance.

B.3.2.14 – Butt Joints: The butt joints used to extend the nose section will use internal sleeves of 0.125-inch wall. There will also be two plug or rosette welds on each tube to help strengthen the butt joint and have over four inches of linear weld.

Project Management

- Tasks complete
 - Frame members cut and bent
 - Frame tacked
 - Fully welded frame
 - Inspection of welds
 - Fabricating gussets
 - Tacking Gussets
 - Fully Welding Gussets
 - Certifying frame welders
 - Prep for finishing (paint or powder coat)
 - Cut nose off of car
 - Fabricate a new nose for the car
 - Install new nose on car
 - Install mounting brackets for components
 - Install suspension
 - Install components
- Modifications completed at competition.
 - Added members at the top of the roll cage to make the four members along the top of the roll cage co-planer.

Front Suspension

Concept Development

It's important to understand what components influence the design of the front suspension. Components that were considered include CV axles, steering column, brake rotors, brake calipers, shocks, and a backing plate to the hub and tire assembly. The biggest mechanical component of front suspension, in terms of size and functionality, is the arms themselves. The arms that were tacked on last year have been pulled off and were changed due to the front of the chassis being extended. Essentially, starting from scratch with the design of the arms, several design considerations were made. Suspension designs are driven by the following:

- Track and wheelbase
- Forces expected to be encountered upon impact
- Max travel (angles and vertical distance)
- Static ride height
- Scrub radius
- Camber
- Caster
- Toe
- KPI (King Pin Inclination)

To understand the importance of these considerations, a lot of research was conducted prior to designing. All these constraints come with some conventional criteria to follow in regard to what

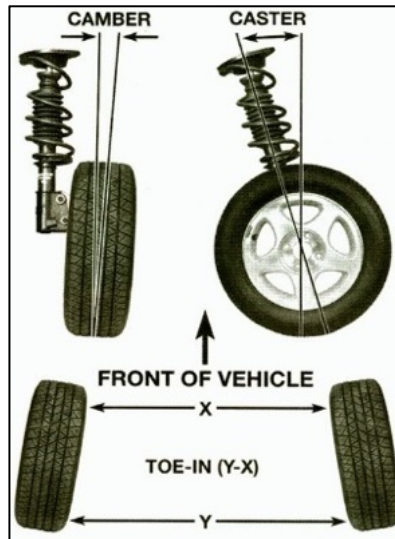


Figure 8: Visual explanation of camber and caster

will produce optimal operating conditions. For example, most cars and ATVs integrate negative camber and positive caster. [3] Figure 8 shows the principle of both camber and caster. A positive caster helps with steering wheel returnability and stability while a negative caster is rarely ever used in any vehicle setting. Having a negative camber will help distribute tire wear and is optimal for keeping traction under the tires as the vehicle encounters bumps.

To control the camber, KPI is a suspension angle that affects the steering and handling of a vehicle (Figure 7). It is the angle between the vertical axis and the steering axis, which is an imaginary line that passes through the upper and lower ball joints of the wheel. KPI also affects the scrub radius, which is the distance between the point where the steering axis intersects the ground and the center of the tire contact patch (Figure 8). A positive scrub radius means that the steering axis of the contact patch. Scrub radius influences the steering effort, feedback, and sensitivity, as well as the tire wear and braking performance. [4]



Figure 9: King Pin Inclination (KPI)

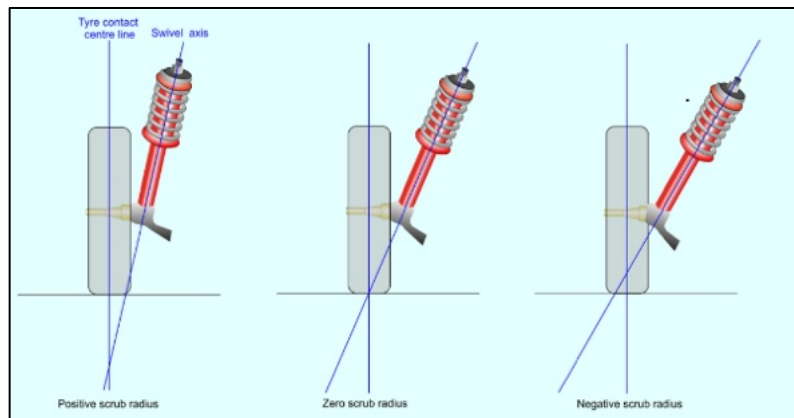


Figure 10: Scrub radius definition

Concept Designs

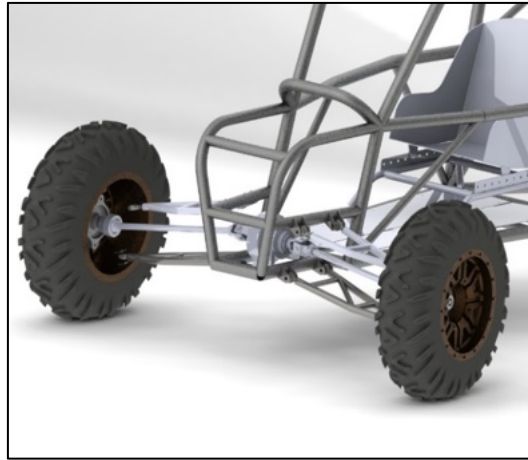


Figure 11: Current front suspension model in the master assembly



Figure 12: Updated front suspension and master assembly

The goal is to have the most optimal design that allows the vehicle to successfully complete challenges when faced with different input scenarios. Specifically for the front suspension, we needed to watch the travel of the suspension arms, steering arm, and CV axles. Problems will arise when a load is encountered, the components max out or end up hitting each other.

The most current CAD model is pictured above in Figure 10. The suspension arm design we chose to integrate is the double wishbone A-arm. This configuration offers better control over the kinematics than a strut-type arrangement. [5] Included in this design is a shorter upper control arm (UCA) and a longer lower control arm (LCA) (Figures 10 and 11 respectively). The UCA will be 1 inch shorter than the LCA. This will induce negative camber upon suspension up travel and positive camber upon suspension down travel. For example, when the vehicle enters a turn positive camber will be gained on the lightly loaded inside wheel, while the heavily loaded outer wheel gains negative camber. This setup will also help control tire wear.

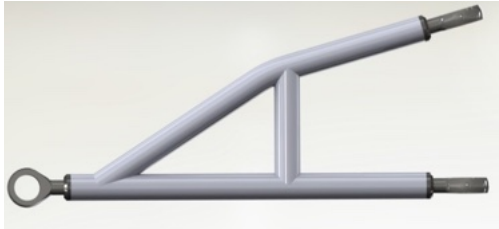


Figure 13: Upper Control Arm (UCA)



Figure 14: Lower Control Arm (LCA)

Currently, the LCA tube length is 18inch and the UCA is 17 inches. With the research done, other design constraints have also been implemented into the CAD model, and the following dimensions will be used going forward.

- Ride Height: 14"
- Wheelbase: 75"
- Track Width: 64"
- Distance between Arms: 4"- 8"

It is important to keep in mind modifications are always taking place. The next steps include doing an FEA analysis on the arms and integrating the lengths with the camber angle.

Tangent to the design of the suspension arms is the backing plate, also known as a hub carrier. The hub carrier is the mounting point for the steering and suspension arms. It also holds the bearing to transfer the rotation from the front differential to the hub and tire assembly. We have decided it is in our best interest to manufacture this part in-house. This is due to the unique design of our vehicle and components.

Components not mentioned located in relation to the front suspension are things that don't have critical design constraints associated with them. These components include items such as brake lines, master cylinder location, and transponder location. Once the driving constraints were defined, integration of other components was easy to apply.

Concept Justification

The first justification for any design would be complying with the rule book. Suspension is one of the few subcomponents that has no restrictions. We, however, were evaluated on what shocks we use and how they are integrated into the suspension design ([C.3.1 – Objective](#)).

Having quality functioning suspension is super important when it comes to performing well at the SAE Baja race. At competition, the dynamic event testing suspension is referred to as the rock crawl. The course is designed to test the limits of each car's suspension. We will expect to encounter large rocks, holes, and hills. Our vehicle's suspension will be put through various heavy loads. It is important to take into consideration all the previously mentioned variables and analyze our step-up under multiple loads.

All the dimensions mentioned in the previous section and constraints that we have elected to be the primary focus for our vehicle have come from extensive research on other schools' vehicles. [6] Teams spend years designing, prototyping, and testing out new designs and have published papers on their research.

Project Management

- Tasks complete
 - Research of suspension arm types
 - Research of backing plates
 - Design of A-arms
 - CAD model of A-arms
 - CAD model of A-arms integrated with master assembly
 - Purchasing hub assemblies
 - FEA analysis of suspension
 - Identifying location for suspension arms
 - Research on driving constraints (ride height, KPI, scrub radius, etc.)
 - Simulation in an online software for suspension
 - Strengthening of A-arms
 - Designing hub carriers
 - Manufacturing of A-arms
 - Install of suspension
 - Integration of shocks
 - Identifying output of current shocks
 - Identifying mounting point
 - Machining and fabrication of hub carriers

Rear Suspension

Concept Development

The design of the rear suspension was centered around many factors and components of the vehicle, like the front suspension. The components with the biggest influence are the CV axles and the rear shocks, with consideration given to the CV axles, shocks, and the backing plate for the wheel hub. The previous and current designs of the rear suspension were based on a trailing arm design. However, the trailing arm that was tacked on last year wasn't complete, so we needed to pull them off and design the system from ground zero. A few of the considerations that needed to be kept in mind during the redesign were:

- Sag length
- Track and wheelbase
- Forces encountered upon impact
- Max travel (angles and vertical distance)
- Static ride height

- Scrub Radius
- Camber and Caster

Concept Designs

The main goal of the rear suspension was to maximize the handling of the vehicle while working with the front suspension to provide a smooth ride. One of the biggest concepts to design around is the ride height; we needed the front and rear ride height to be within 1”-2” of each other.

The trailing arm design was chosen because it gives the most control for an off-road setting. The trailing arm design consists of a trailing arm (Fig. 10-12) that connects to the chassis and holds the mounts for the shocks, rear toe links (which attach the trailing arm to the rear of the chassis) and backing plate (also known as the hub carrier). The rear components were set up to mirror the camber of the front suspension, where positive camber is achieved through suspension down travel and negative camber for suspension up travel.



Figure 15: Orthogonal View of Trailing Arm

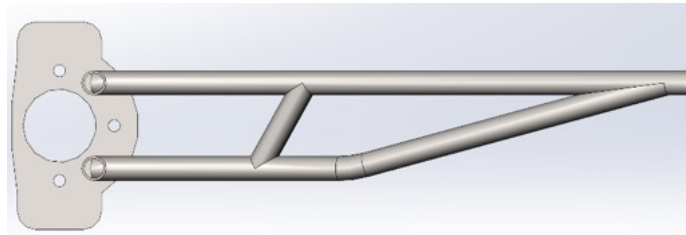


Figure 16: Front View of Trailing Arm

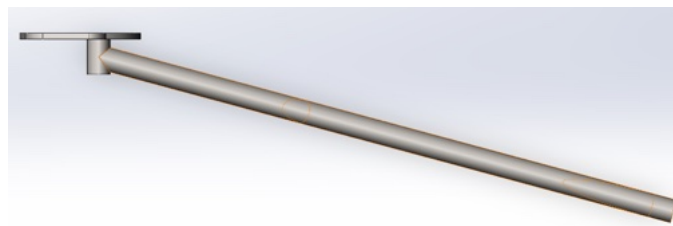


Figure 17: Top View of Trailing Arm

The current dimensions of the trailing arm set the track distance at 64", with a wheelbase of 75", mirroring that of the front of the car. The arm also sets the ride height roughly equal to that of the front, which is 14". Consideration had to be given to other important angles that are included in suspension design, such as the scrub and KPI angles. Please refer to the concept development segment under the front suspension section for more details regarding these specs.

While not as important as the front suspension, close attention was given to the CV axles to make sure there were no collisions with the rest of the rear suspension or chassis. The max compression distances were used to make sure there would be no component collisions during full compression or extension of the shocks.

The current design of the rear suspension allows for a theoretically optimal ride; however, changes occurred after further analysis. The next step in the process was FEA testing and testing/prototyping following manufacturing. There may be adjustments that needed to be made to change certain constraints that our preliminary design didn't consider.

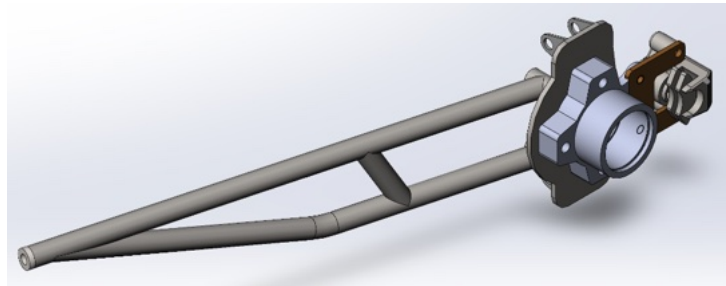


Figure 18: Trailing Arm with Hub and Mounting Tabs

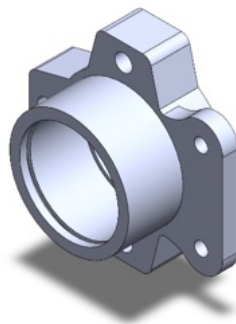


Figure 19: Rear Hub Carrier

As with the front, a backing plate needs to be integrated with the trailing arm. This plate (or hub carrier) is where the tire assembly connects to the suspension assembly and mounts directly to the trailing arm. As compared to the front, the rear hub carrier only needs to house the bearing that will redirect the rotation of the rear differential to the tires and the mount for braking. These front and rear plates were manufactured simultaneously in-house, as opposed to outsourcing

them. Just like the front hub carrier, disk brakes mount onto the rear hub. This allows easier brake mounting, as all brake lines will mount to the same point on the pedal box.

Once the trailing arm was fabricated and mounted onto the chassis, we manufactured and fitted the rear hub onto the arm. Once this was complete, we could install the brakes and shocks, and put the tires on, giving us our final manufactured prototype (Fig. 15).

After designing the rear suspension in CAD, SAE Baja required testing to be done on each subsystem. While we were unable to gather experimental data on the suspension prior to the competition, we completed FEA on the suspension subsystems of the car.

As can be seen below (Fig. 20, 21), the FEA shows that there were no major concerns. The FEA was done with an offset load using forces exceeding 1,000 lbf. The FEA showed that there were no major areas of concern. In the future, more complicated force placements could be utilized to get a more detailed result, but the rear suspension withstood competition.

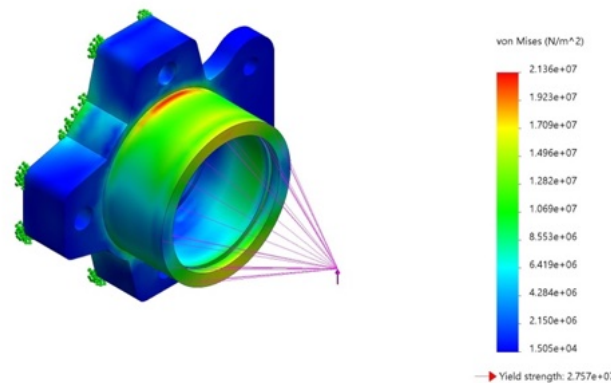


Figure 20: Rear Hub Carrier FEA

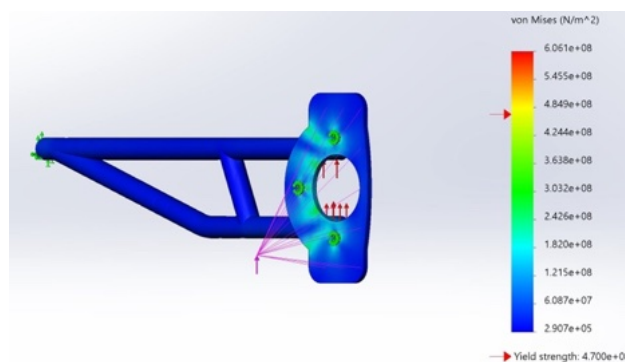


Figure 21: Rear Trailing Arm FEA

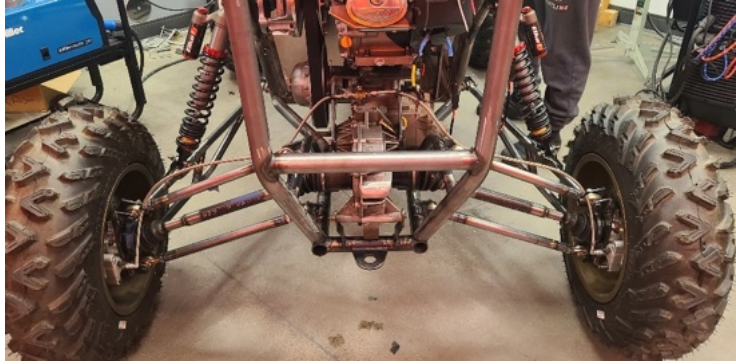


Figure 22: Rear view of rear suspension

Testing of the entire rear suspension subsystem took place at the SAE Baja competition in California this April, where adjustments were made to the toe and camber of the wheels for proper alignment. With limited running, the car showed no signs of concern. More testing and operation of the vehicle will be required to unearth potential problems with the design.

Concept Justification

When it comes to SAE, most subsystems require strict compliance with the rulebook: however, no such restrictions were present for the suspension. All justifications in the design were made purely for the performance of the vehicle. The shocks that are used and their integration will be assessed at the competition ([C.3.1 – Objective](#)).

Regarding the SAE Baja competition, suspension is one of the most important subsystems. The front and rear suspension must work in tandem to provide the best performance possible. The rock crawl event, alongside the other dynamic and specialty events, has a heavy reliance on the suspension system of the car. This is why all our preliminary design and testing yielded the strongest, most reliable suspension system available. As our car is on its first fully designed prototype, it is very important to keep the various constraints in mind while furthering the design. Testing of the vehicle was also a very critical task in the completion of this project, and analysis of said testing led to the best version of the suspension we could have provided for competition.

Project Management

- Tasks complete
 - Material required to complete suspension design is already in-house
 - Research on ATV and Baja rear suspension systems
 - Design of trailing arm
 - CAD model of trailing arm
 - CAD model of rear hub carrier
 - Assembly of entire rear subsystem
 - Integration of assembly in master CAD model
 - Manufacturing of trailing arm
 - Manufacturing of rear hub carrier

- Mounting of trailing arm, hub carrier, shocks, and wheels
- Tasks in progress
 - Testing of rear suspension in California

Drivetrain

Concept Development

The major components of the drivetrain include the engine, CV transmission, driveshaft, and front differential. Subcomponents that the drivetrain group is responsible for include working with suspension on the CV axles, all mounts and tabs associated with major drivetrain components, a tunnel cover for the driveshaft, and all HROE shields and finger guards for the driveshaft. Most of the mounts in the rear of the chassis were completed by the previous year's team. However, none of the CAD models associated with them were created. This left our team to focus on converting what they had done on the chassis to a working CAD model. Although the rear was mostly finished, the positioning of the front differential has been changed due to limited space in the toe box to include room for the steering components, front differential, and pedal assembly. The following design considerations have heavily driven the current model:

Rear design considerations (engine & transmission):

- Existing transmission output location
- Keep the existing mounts for the engine and transmission
 - These mounts were later modified to thicker steel as well as additions to adequately support the weight and force outputs of our engine

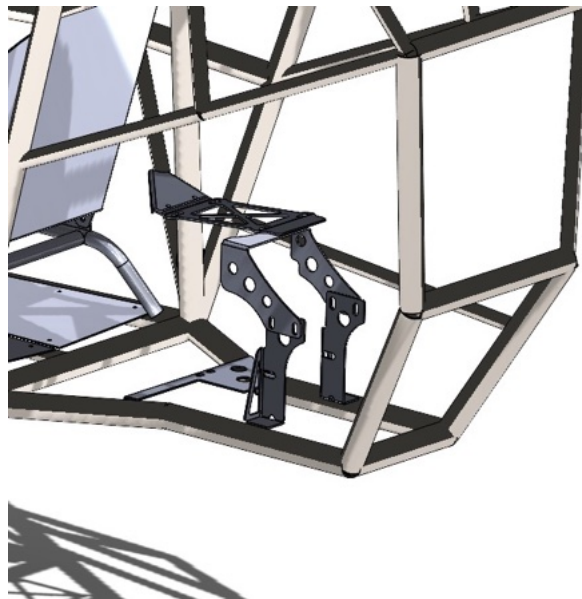


Figure 23: Engine and transmission mounts

Driveshaft design considerations:

- What is the maximum U-joint angle under the seat in order for the driveshaft to be once continuous shaft from the CVT output to the front differential input
- Mounting locations for support bearings, this was later removed as we changed our design from two driveshafts with a u-joint to just one direct shaft, if future designs include a broken driveshaft, you will need to include a support bearing for each section of the driveshaft

Front differential design considerations:

- Axle output centered with wheel and suspension components; this determined our mounting height for the front differential
- Driveshaft input height achievable with existing bottom panels

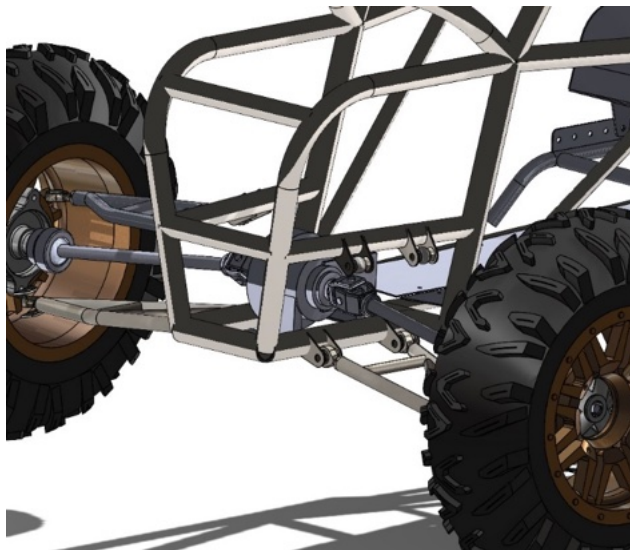


Figure 24: Isometric view of differential and suspension components

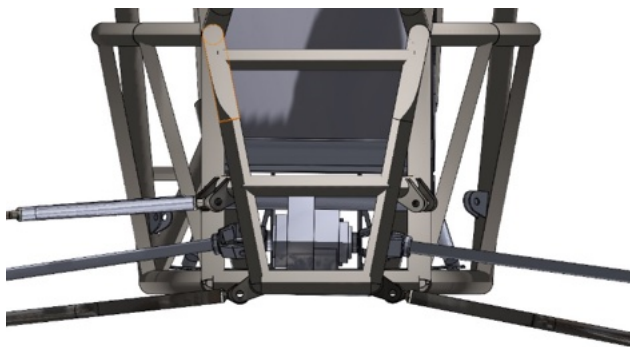


Figure 25: Front view of differential showing axle output angles and suspension angles

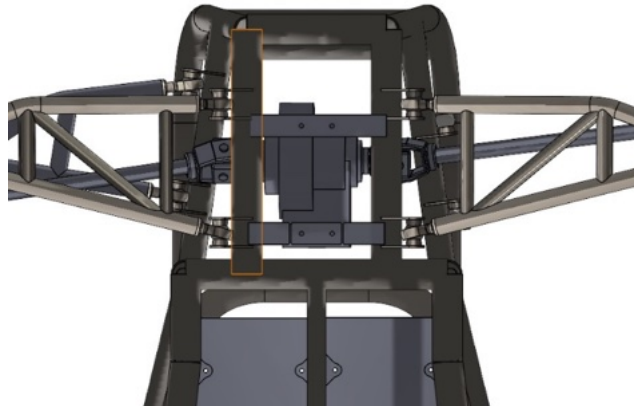


Figure 26: Bottom view of differential showing increased tow box area



Figure 27: Initial driveshaft location plan

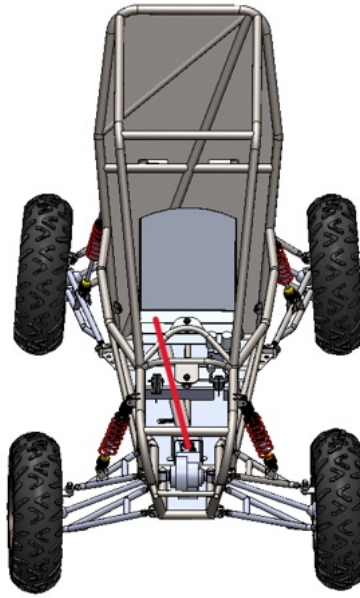


Figure 28: Final driveshaft plan

Concept Designs

The designs we have created for drivetrain components have been focused on the goal of keeping as many components currently installed on the chassis as possible included in the final design. This is because we are able to make minor modifications to include our designs, as well as the fact that much of the mounting hardware used to install the drivetrain has been fully tig welded onto the frame. The following lists in the project management section of the drivetrain show what designs and concepts were finalized for the drivetrain.

All new concept designs were mainly focused on adhering to the rules stated in the SAE rulebook. Many of the drivetrain rules stated minimum material specs required to use, as well as where guards and finger guards are required. This heavily guided the drivetrain design and did not allow for many large changes. Many designs we used were the simplest way to achieve the desired outcome while also adhering to the rules.

Concept Justification

After we determined the front differential's location, we began torque calculations on the driveshaft and axles. This verifies our choice of driveshaft material and justifies our selection of CV axles. All other concepts and designs for the drivetrain were heavily influenced by the SAE Baja rulebook and integration with other sub-systems. These designs include the size and positioning of our driveshaft tunnel, the front differential location, and the rear axle outputs.

Once again, much of our justification for these designs was based upon the requirements stated in the rules.

Project Management

- Tasks complete
 - Engine and transmission mounts installed
 - Engine installed
 - CV transmission installed
 - Front differential relocation
 - CV axle lengths being determined
 - Driveshaft material selection
 - Gas tank and gas lines installed
 - Collaborate with suspension on brake locations
 - Wiring for CV transmission, engine, and front differential
 - All drivetrain shielding and guards manufactured and installed
 - All drivetrain components have been completed

Steering

Concept Development

The following paragraphs will run you through the process of how the steering components got to their final concept design.

The previous year's team had ordered most of the components for the steering assembly. We had to sift through the parts and put the components in SOLIDWORKS. Once we had the components in SOLIDWORKS, we used the parts in the senior capstone bay to get a rough idea of how to plan our steering.

The main goal of figuring out the steering was to determine the placement of the rack and pinion. Since this vehicle is 4-wheel drive, the rack and pinion location were altered due to the front differential. The front differential is a relatively new addition to the Baja competition rules. With the addition of the 4-wheel drive component, the area to work around in the front (tow) box was limited. This, in turn, was a big challenge for designing a mount for the rack and pinion.

Our initial design was to mount the rack and pinion above the suspension for the ease of not moving the differential. This initial design was redesigned after we came up with the issue of the tie rod and shocks buckling. After we determined that our first design was no longer practical, we had a brainstorming session with Greg, Eric, and the team. We ended up getting the idea to move the front differential backward, extend the front tow box, and put the rack and pinion in the front of the vehicle. This idea of a front-locating steering system was continued, implemented, and designed into what we now call our final steering concept. So, ultimately, the extension of the tow box allowed us to complete our steering. The basic components of the steering assembly are shown in *Figure 25* below.



Figure 29: Steering Assembly in SOLIDWORKS (minus U-joints)

Concept Designs

Steering is one of the under looked systems in the vehicle. It is the only thing that allows it to stay on the road. Our goal was to design a steering system that could maneuver all the other components in the front along with having the proper Ackerman. Ackerman steering is the geometry of the tires in both front wheels when you turn the steering wheel. The figure below shows a type of steering that we will be basing our wheels on. The model shows how the tires will behave based on the rack and pinion being in the middle of the wheel rim. Our final steering bracket will have the tie rod attachment in the front section of the wheel rim. We will have to flip the rack and pinion to take the proper action we need. With the normal orientation, the bolt holes facing the driver, the car's steering would be opposite of what we would like. So, with the opposite orientation, we get the proper steering, which is turning the steering wheel left to go left.

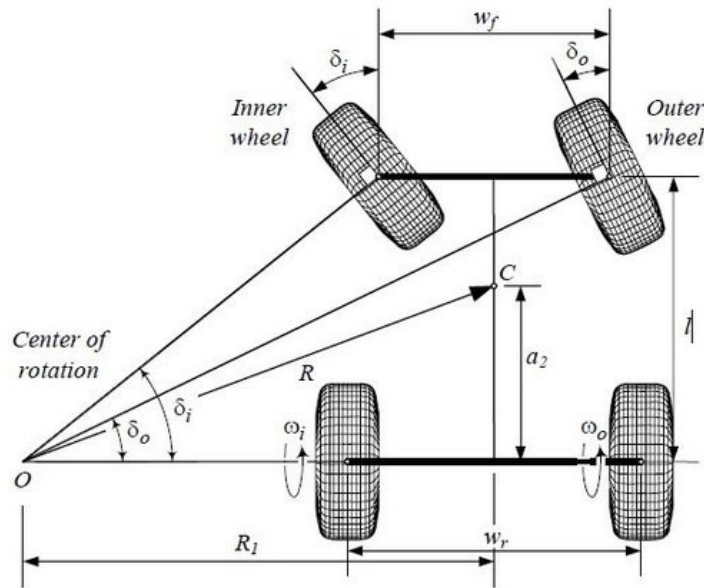


Figure 30: The Angles and Lengths Involving Ackerman Steering

The steering system concept is finished. The team has added the U-joints for the steering column and the last brace for the steering column into the vehicle. Looking toward the future, upcoming classes should continue to implement a takeout design, meaning it is easy to remove for repairs.



Figure 31: Implemented Rack and Pinion

The rack and pinion are fully implemented into the master assembly. We are running with reverse Ackerman steering, meaning that tie rods are attached to the front side of the wheels. Our reasoning for this setup is due to the lack of space behind the differential. The extension of the tow box has allowed enough room for the rack and pinion to comfortably fit and be in line with the suspension arms. We initially ran into some issues regarding the buckling and eventual failure of the tie rods that would occur if our car was in complete compression. All these issues have been addressed.

Concept Justification

Our concept justification is the Baja SAE Rulebook. If everything is designed to comply with section B.8.7 Shielding for Legs and Feet, the steering design is up to the Baja team. One thing that is not a justification, but a recommendation is quick disconnects or accessible fasteners. Most of our calculations and measurements were determined based on either other Baja teams or modeled in CAD.

Project Management

- Tasks complete
 - Full steering assembly designed with rack and pinion mount point
 - Spacing and ergonomics of steering
 - Steering assembly modeled and mated into master assembly
 - Finding exact U-joint dimensions and putting them into CAD
 - Ordering all steering components
 - Steering wheel attachment
 - Bracket for steering column
 - Installing steering components
 - Installing steering brackets
 - Fully installed and tested steering system

Project Tasks

Our team will complete the following tasks to build a vehicle capable of racing in April of 2024. Note that most tasks are happening simultaneously.

Task 1. Finalize CAD Model

Description: Generate a fully-defined Computer-Aided Design (CAD) Model that replicates our final design.

Task Leads: Tyler Baker, Teag Parker

Timeline: January 8th, 2024

Budget (Hypothetical): \$0

Deliverable: Completed CAD model of the Baja car.

Subtasks:

- *Build the sub-systems:*
 - *Frame*
 - *Drivetrain*
 - *Suspension*
 - *Steering*

- Engine/transmission
- Assemble sub-systems in the master assembly
- Constrain sub-systems in master assembly

After completion notes on this task: This task continued to be ongoing throughout the entirety of this project. The biggest challenges with having a complete CAD model of the car were the organization due to many people contributing models to the master assembly and constraining/mating the master assembly. The best way to ensure the CAD model was organized was to be diligent about uploading models from personal computers onto the main Baja computer. Organizing files by subsystem within the master assembly also proved useful when trying to find parts. Another recommendation we have is to include all nuts/bolts and any small off-the-shelf parts in the assemblies, this makes finding which bolt needs to be used much easier, as well as ordering new bolts if necessary.

Task 2. Finalize Frame Fabrication

Description: Finish welding the frame of the vehicle.

Task Leads: Jayden Kuhl

Timeline: January 2023

Budget (Hypothetical): \$0

Deliverable: A 100% hard-welded frame.

Subtasks:

- Weld frame members
 - Check on the current roll cage for completeness.
- Design suspension mounting points on the frame
 - Weld the mounts for suspension.
- Design steering column
 - Weld the mounts for steering.
- Design drive train
 - Weld mounts for drivetrain.
 - Design suspension components (A-Arms, drag links, etc.).
 - Weld suspension components.

Many of these tasks depended on other tasks. Direct communication and collaboration with other task leaders will be necessary.

After completion notes on this task: This task's biggest challenge for us had to do with the design of the frame and making it work with our front suspension as well as having room for all drivetrain and steering components in the toe box. Although this was mentioned earlier we will reiterate it here, our recommendation for the frame design is to design the front members around the style of suspension you want to use, as well as make sure you have enough room to include your front differential, steering components, and pedal box plans. This would have saved us

many hours of work as we would not have needed to extend the toe box or have struggled as much with the design of our front suspension.

Task 3. Finalize Suspension Geometry and Fabrication

Description: Run calculations to ensure the proper function of the suspension system. Fabricate and assemble all remaining suspension components.

Task Leads: Shelby Olson

Timeline: 1 ½ Months

Budget (Hypothetical): ~ \$500

Deliverable: Completed suspension installed on the vehicle.

Subtasks:

- *Update suspension CAD model.*
- *Fabricate tabs and other parts for suspension.*
- *Prep frame for welding.*
- *Tack the mounting tabs, marking out the location of the suspension.*
- *Approve for full welding and send off to Jayden Kuhl.*

The front suspension will be designed to be double wishbone-A-arms, while the rear suspension is a trailing arm attached to the middle of the chassis and two toe links attached to the rear. The CAD model and FEA analysis will be used to finalize the placement of the arms and angles of the shocks.

After completion notes on this task: This task's main recommendations have been stated before, as well as in the previous task. Design the front suspension first and the frame around that.

Task 4. Design/Fabricate Backing Plates

Description: Moving forward we will no longer be utilizing portal axles as our backing plates. This decision was made to reduce the complexity of the car's design and reduce the risk associated with portal axle failure during testing or competition. Moving forward we will be designing and fabricating our own hub carriers to allow for adjustability is our suspension characteristics.

Task Leads: Tyler Baker

Timeline: 3 Months

Budget (Hypothetical): ~ \$300 (aluminum stock to machine).

Deliverable: four completed hub carriers (Front(left and right), Back(left and right)).

Subtasks:

- *Research similar designs*
- *Design hub carriers*
 - *Front and Rear*
- *Verify interface with CV axles and hubs*
- *Create machining paths*
- *Fabricate/machine hub carriers*
- *Install hub carriers on car*
- *Mount brakes and CV axles*

After completion notes on this task: After completion of the hub carriers and seeing them perform at competition we were overall satisfied with the performance. For as much running as we did, we had no issues with suspension bolts or hub bearings. Future recommendations would be to optimize this component for weight and suspension/steering performance. Reducing the weight of this component would achieve greater bump performance through reduced unsprung mass. Utilizing CAD and computer models for suspension and steering geometry at the same time would greatly improve performance all around because these systems directly influence each other in all scenarios.

Task 5. Preliminary Testing of Vehicle

Description: Bringing the vehicle to a location to test its maneuverability, functionality, and durability.

Task Leads: Tyler, Shelby

Timeline: 4 Months

Budget (Hypothetical): \$0

Deliverable: Finding and repairing/redesigning points of failure and improvement.

Subtasks:

- *Road test car.*
- *Trail test car.*
- *Redesign points of failure.*
- *Continue to test and redesign as needed.*

After completion notes on this task: Due to the design and manufacturing of our vehicle taking longer than expected we were unable to truly test our vehicle. From what little testing we did conduct we still learned a lot. Our recommendations for this task are to ensure you have enough time to truly test. We found many issues many of which arose from the engine placement and pedal box. Every aspect of each subsystem is critical for getting the vehicle to run so one small error in any subsystem could entirely stop the vehicle from driving. Another thing to take note of

is that even though the CAD model runs perfectly that doesn't mean it will all be manufactured and assembled perfectly. Beyond that even if the vehicle looks good and performs well when sitting in the welding bay stationary, we still found a variety of problems in our quick tests outside while driving the vehicle.

Task 6. Redesign and Rebuild Based on Testing

Description: After testing our initial vehicle, we plan to find many failures and points of improvement, which we will address here.

Task Leads: ALL

Timeline: TBD based on testing.

Budget (Hypothetical): TBD based on testing.

Deliverable: Fixing and upgrading all points of failure.

Subtasks: TBD based on testing.

After completion notes on this task: This task is similar to what was mentioned in the previous task. The biggest recommendation we have for this is to ensure you have enough time to truly test the vehicle, in order to diagnose problems and have the time to adequately fix them.

Task 7. Train Driver

Description: Determine who our driver will be and train them to compete to the best of their abilities given the vehicle we manufacture.

Task Leads: ALL

Timeline: 1 ½ Months

Budget (Hypothetical): Cost of gas to train ~ \$20

Deliverable: A trained driver ready to compete in the competition.

Subtasks: Conduct time trials to determine our driver.

After completion notes on this task: With the vehicle taking longer than expected to complete, we did not have time to train any driver. Looking back at competition this task wasn't of great significance. Our car wasn't fast enough to compete with any of the top runners and the need for fast lap times was insignificant. This may be useful in later years, given enough time, when we have a fast car that can compete against other schools in dynamic events.

Task 8. Bring the Vehicle to Race in Gorman, CA

Description: After all tasks are completed, we will travel to California to compete in the national competition with opposing universities.

Task Leads: Tyler, Jayden, Shelby

Timeline: April

Budget (Hypothetical): ~ \$3,000

Deliverable: Our completed vehicle.

Subtasks: All tasks must be completed. Securing a trailer for transportation, and truck for pulling said trailer. Booking hotel rooms for the stay in Gorman.

After completion notes on this task: Learning outcomes from this task are summarized in the “Competition Takeaways” section a few pages after this one.

Project Deliverables

Deliverables of this project will be broken down into two distinct categories. The first is our senior design deliverables, which will be produced in the context of the Senior Capstone class. The second is our SAE-specific deliverables, which contribute to obtaining points in competition in California.

We will deliver the following:

- Senior Capstone Class
 - Written Preliminary Design Review report.
 - Preliminary Design Review PowerPoint presentation and accompanying presentation.
 - Critical Design Review.
 - Final written plan and project presentation.
- SAE International
 - Business Presentation Plan.
 - Budget and Cost Overview.
 - Design Documents.
 - Technical Documents.
- Drivable car with the ability to compete in the competition.

Timeline

The full project started in October 2023 and was completed by the third week of April 2024. The deliverables timeline for the project is as follows:

[Table Link](#)

Table 7: Project Timeline 1

#	TASKS	2023																	
		August		September		October		November		December		January		February		March		April	
		1st Half	2nd Half	1st Half	2nd Half	1st Half	2nd Half	1st Half	2nd Half	1st Half	2nd Half	1st Half	2nd Half	1st Half	2nd Half	1st Half	2nd Half	1st Half	2nd Half
1	Finalize CAD Model																		
1.1	Finish Frame CAD Model																		
1.2	Finish Suspension CAD Model																		
1.3	Finish Drivetrain CAD Model																		
1.4	Finish Portals CAD Model																		
1.5	Finish CVT CAD Model																		
2	Finalize Frame Fabrication																		
2.1	Obtain Certified Welder																		
2.2	Finish Welding Frame																		
3	Finalize Suspension Geometry and Fabrication																		
4	Fabricate Portals																		
5	Begin Testing Vehicle																		
6	Redesign/Rebuild Based On Testing																		
7	Train Driver																		
7.1	Time Trials																		
7.2	Train Driver At Hartmans																		
8	GORMAN CALIFORNIA																		
8.1	Get to Gorman California																		

Competition Takeaways

The SAE Baja California competition was attended by senior capstone members Jayden Kuhl, Shelby Olson, and Tyler Baker as well as four junior and one sophomore club members in April of 2024. This was the first group ever to represent Western Colorado University at an SAE event. Below is a summary of the event itself, takeaways from the event, and future recommendations for the Senior Design class, club, and participation in SAE events.



Figure 3213: Tech Inspection Passing Stickers

Competition Agenda

SAE Baja hosts three annual competitions in the US, two in the spring and one in the fall. The competition we attended was held in Gorman, CA from April 25th - 28th. Every competition is different, but for the most part, it will follow the same schedule.

Day 1: Loading Day and Technical Inspection

Teams can set up their pit areas and unload vehicles. Teams are also able to get in line for engine tech. If any repairs or last-minute things need to be done this is the day to complete all of those.

Day 2: Technical Inspection and DRB Presentations

A lot is going on this day. Teams can get through engine tech if they didn't the day before. Also, general tech inspection opens for teams on a lottery-based system after the top ten teams get through. 4WD check opens and can be done at any time after waiting in line. Dynamic brake checks are open and can be completed after general tech inspection. Teams are also assigned a time to present their DRB to a panel of judges.

Day 3: Technical Inspection and Dynamic Events

For teams that didn't get through general tech inspections, this is the last day to do so. Inspections are on a first come first serve basis. This is also the start of the dynamic events including the hill climb, acceleration, S&T course, and maneuverability. Teams can complete them in any order they wish. Acceleration results are used to set the grid for the endurance race.

Day 4: Endurance Race and Banquet

The final day is held just for the 4-hour endurance race. Teams have the morning to make any last-minute adjustments and get their fuel and paddock pits ready for the race. The race is held, and teams can fix their cars as the race continues. Results are determined by how many laps each car completes. The day finishes with a banquet and award ceremonies.

Technical Inspections

Full vehicle inspection is broken up into 4 different categories: Tech inspection, 4x4 check, dynamic brake check, and engine check. You are not allowed to compete in any dynamic events until all four

inspections are passed. Another rule is you may not drive your Baja vehicle around the site until you gain all four stickers. Below is a breakdown of what each inspection entails.

General Tech Inspection

The biggest and most comprehensive inspection is the general tech inspection. Here SAE officials inspect the entirety of the vehicle using a checklist that goes over everything in the rulebook. The frame and chassis are inspected for weld quality, required bracing members, and no tube or weld defects. Next, they move on to check every subsystem for required parts and safety regulations. The biggest of which is HROE guarding and finger protection of spinning parts. They will also test safety equipment such as fire extinguishers, helmets, fire suits, goggles and diagnose driver egress of the vehicle. Only two members of the team are allowed in the tent for general tech inspection. This is the longest and most time-consuming part of tech especially if you have to go through it multiple times. Try to get in as early as possible and get through first try. Not passing tech in a reasonable time deducts points from your overall competition score.

4x4 check

Beginning in the 2021 season 4-wheel drive / all-wheel drive became a requirement. Assuming you have tested 4-wheel drive prior this is the easiest inspection to pass. It rarely had a line to wait in and took about 5 minutes to complete. For this inspection they put your rear wheels on rollers and instruct you to drive forward, proving 4-wheel engagement.

This inspection can be completed after engine tech prior to Tech Inspection. Our recommendation is to get it done and out of the way as soon as possible.



Figure 3316: 4 x 4 check

Dynamic brake check

The dynamic brake inspection must be completed in order to participate in dynamic events. After passing general tech you can demonstrate the vehicle's braking ability where all 4 wheels must lock upon hitting the brakes. Failure to pass could require rebleeding of the brakes. If done correctly this should only take about 5 minutes.



Figure 34: Dynamic Brake Check

Engine check

Engine tech is the first step in getting through tech. Here Kohler employees inspect your engine to make sure you are running the correct engine and restrictor plate. They ask you to disconnect the CVT and make sure all adjustment points are easy to access. From there they check and adjust your idle and max engine rpm. Along with the engine specs they ensure that you can reach full throttle of the engine with a foot actuator and cable.

Design Briefing Review Presentation

Follow the DBR packet as a template. The judge's scoring sheet is based on this template, and it allows for the most points. The presentation only lasted 20 minutes, and it went by fast. Bring a minimum of four people to the presentation; each person should be comfortable presenting on one section of the DBR.

Bring graphics to help earn more points for testing and CAD. Guide the judges through your goals, design process, testing, and verification. Judges are trying to give you as many points as possible so do your best to touch on everything. If they have questions their aim is to try and get you more points.

Packing List

All the below items are items that we used and/or wished we had while at competition in California.

- Pop-up tent
 - Bring big stakes to hold the tent down
- Cooler
- Snacks/water
- Sunscreen
- Camp chairs (lots of waiting)
- Fuel for car
 - Ethanol-free or premium is best for the carburetor on the car and generator
- Generator
 - Make sure it is powerful enough to run all of the corded equipment and has the correct plug for the welder
 - Welder generators are a nice backup when the MIG machine does not work or having a suitcase MIG with the welder generator allows for the smoothest operation
- Welder(s)
 - You are out in the elements, flux core MIG helps with shielding when gas would be blown away, stick welding with small diameter electrodes is also an option when all else fails
- Grinders
 - Cut off disks
 - Grinding disks
 - Flap disks
- Air compressor
- Small hand tools
 - Pliers, hammers, dead blow, punch, screwdrivers, ect
- Metric and SAE Wrench set
- Metric and SAE ratchet set
- ½ Impact
- 3/8 Impact
- First aid kit
- Sheet metal
 - Works nicely as a work table that transports easily
- Extra primary and secondary tubing
- Extra odd-ball size tubing
- Brake fittings
- Saw horses
- Electrical fittings
- Wire

- Crimpers
- Strippers
- Soldering iron
- Aluminum tape
- Fluids for the car
 - Transmission, brakes, engine
- Wagon or offroad rolling tool chest
- Bins for driver safety equipment

Competition Culture

The competition has a very strong collaboration culture. Everybody that is there knows the time and effort put into building a Baja car. Most teams are more than happy to talk about their car's design and help you when you are in a bind. There are teams that have machine shops in their trailers; other teams have printers. Making friends at competition is easy and critical to a successful year. Help goes both ways; give it when and where you can, and don't be afraid to ask for supplies when you find out you do not have something.

Communication

Short-distance communication via radios or walkie-talkies can streamline competition. Being able to communicate with different team members across the competition site can help when repairs need to be made or if a tool needs to be brought down to where the car is. Radio communication with the driver can help during the endurance race. If and when something breaks on the car, the driver can radio it over to the team to start getting the pits ready to make the repair and have a person ready to walk the car out of the race area. If there is a black flag on the car the driver is told the reason for the black flag, again allowing the team to get ready to fix the issue on the car before the car is off the track.

General Takeaways

Everybody at the competition is rooting for as many cars to make it to the endurance race as possible. Teams are always ready to help other teams out, and judges are willing to let you change items as they inspect the car, given that they are little things.

Always have a plan A and a plan B at competition. Being away from a fully furnished shop can introduce unexpected problems. Being able to adapt and make use of tools and materials in ways they are not typically used is critical to being able to modify the car as needed in competition.

While at competition, you spend all day outside in any range of elements. The California trip was expected to be sunny and warm. What we got was cold winds, blowing sand, and a couple rainy days. Be prepared to be outside for hours on end in less-than-ideal conditions.

Triple-check everything. Understanding the rule book thoroughly. Even when you think you understand everything, it is possible that the judges have a different interpretation of the rules. Make sure everything is Baja/SAE compliant. Lights, engines, and hardware should be double and triple-checked to ensure they meet the SAE requirements.

Qualifications

Spencer Romig – Student, Mechanical Engineer

I am responsible for assembling the SAE BAJA vehicle and participating in the vehicle's manufacturing. With my background in manufacturing shops and a big interest in the automotive/marine industry, I will be able to provide the club with another perspective on the project.



Jayden Kuhl – Student, Mechanical Engineer

I am responsible for leading the fabrication of the SAE BAJA car and monitoring the club's finances. Drawing from my experience of running my own business, I will be able to guide the club on the wise use of funds. I have also gained fabrication experience from previous jobs and shop time at the university.



Teag Parker – Student, Mechanical Engineer

I am responsible for compiling existing CAD models and maintaining the organization of the CAD model as more is added throughout the project. I will also focus my manufacturing and research efforts on the drivetrain subsystem for this project. My background and qualifications for this project mainly come from my passion for the automotive industry and my experience last year as a member of the Baja SAE club.



Tyler Baker – Student, Mechanical Engineer

As a member of this project, I will be mainly responsible for the project management of the club and senior design team and the design and manufacturing of the hub carriers. My background as a TA in the maker spaces provided me with skills in machining and more advanced manufacturing techniques that will be necessary for the completion of the car. As the leader of several previously similar projects, I will lead this senior design team to collaborate on each subassembly of the SAE Baja car and communicate their designs with the SAE Baja club.



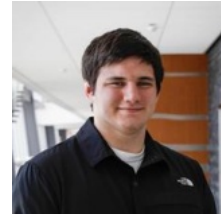
Shelby Olson – Student, Mechanical Engineer

My contribution to this project will stem from my background in automotive technology. My experience servicing motor vehicles will be vital in achieving a race-ready running vehicle by April 2024. I will be taking the lead on each component design and ensuring we operate within the competition's rules and regulations.



Gable Watson – Student, Mechanical Engineer

As a team member, my main contribution will be assisting with the assembly of the vehicle and manufacturing parts. My manufacturing and team communication background will provide the club with proper information distribution and assistance with developing subsystems.



Deagen Fahrney – Student, Mechanical Engineer

My main contribution to this project will be in the manufacturing and design of the vehicle. I will primarily contribute to the project through experience in the design and manufacturing process. Although new to this project, I have worked extensively in a team setting, which will help the project flow. This will help the development and design of the subsystems of the car that will lead to its completion.



Appendix A – Results from Stakeholder Mapping

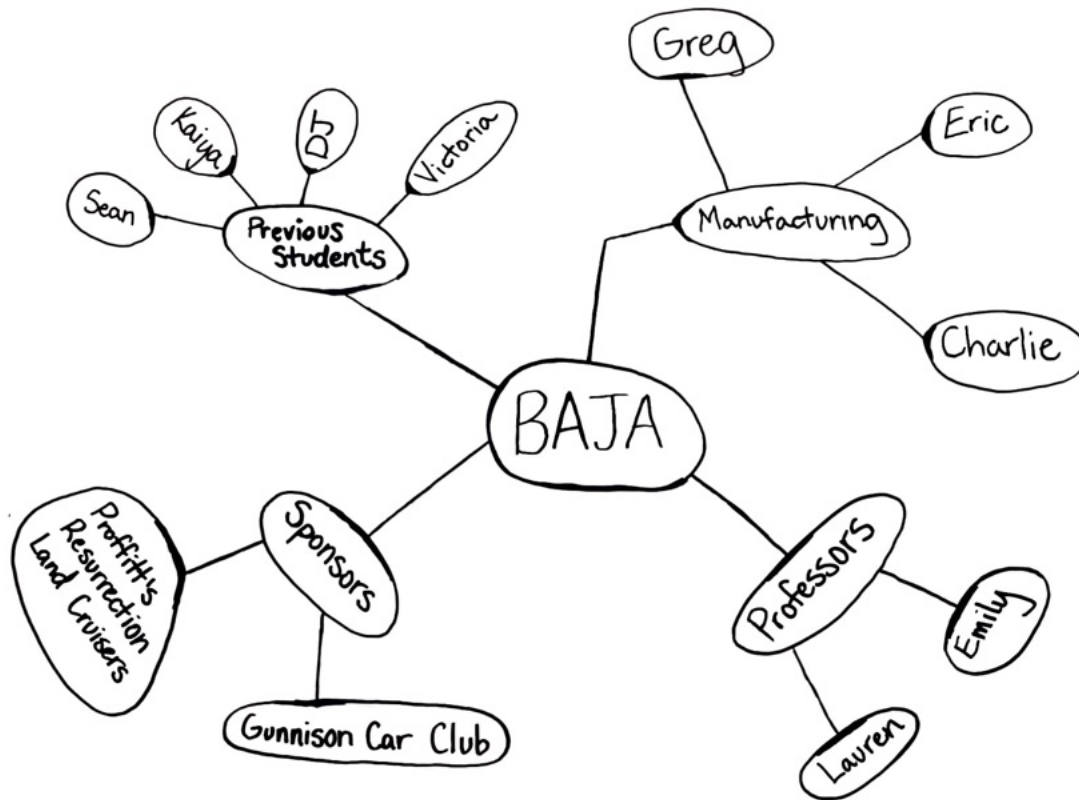


Fig. 3: Stakeholder Map

Appendix B – Results from Stakeholder Interviews

B.1 - Jayden's Interview with Zak Trafton

After a short interview with the president of the Gunnison Car Club and part owner of H&H Towing, Zak Trafton, I understood more of what the community would like to see coming out of the BAJA project. Zak's first point was producing a “competitive entry with innovative design elements.” Zak hopes to see something unique on our car, like portals or other design elements. He also wanted to see something that caught bystanders' attention: a clean paint or graphics job that makes the car look complete and cared for. The next point Zak and I discussed was regarding features integrated into the Baja car. Zak mostly talked about adaptive suspension. He wanted to see something stiffer on a smoother course and softer for rugged courses, like jumping logs and big drops. He also touched on the portals allowing “instant ground clearance” without other drawbacks from lifts, such as CV and A-arm alignment. While discussing the team structure, Zak reaffirmed our current structure of having a project manager overseeing the entire project with smaller teams working on major subsystems. This allows for a streamlined workflow, similar to how the H&H shop works. Finally, we discussed ordering parts and combating long lead times on products. The first way to cut lead times would be to upcycle products locally. The second way we can cut the lead times is to use readily available products that are more of an off-the-shelf product instead of custom products that are manufactured or made to order.

B.2 - Teag's Interview with Lauren Cooper

This interview did not involve going over design requirements and suggestions, so I will not talk about how this interview will affect our design but how it will change how we work as a team. The main takeaways I gained from this interview with Lauren Cooper were related to our performance in this class, what that looked like, what it looked like last year, and how we can achieve our goal grade and success. A main point discussed a lot was how last year's team handled criticism from each other and how big egos played a role in a less successful project. Lauren emphasized her advice that we should maintain a cooperative team willing to compromise on things we disagree on instead of having one person who controls and tries to dictate all project decisions. I also asked about how our success would change in the class, whether we get a functioning car, and if going or not going to the national competitions would affect our grades and success in the class. Her main point about this was that a good goal would be to have a running/ drivable car to take out into the valley and test. Whether or not we go to the competition and our success plays no role in how our grades in the class turn out. Also mentioned was if we have any “due dates” for SAE Baja material, we should be in close communication with Lauren and go from there about what will be expected of us for Capstone deliverables. Lauren also mentioned that portals are a very cool idea; however, they may take longer than expected. This is not suggesting we don't do them; it is just warning us.

B.3 - Tyler's Interview with Eric Ortolano

Several key insights were brought up repeatedly by interviewing Eric Ortolano, one of the makerspace shop managers. The first, and arguably most important, is communication. One of the struggles of the previous group was that they didn't communicate enough between subsystems, which led to systems not meshing or not considering the entire project with all the interfacing parts. Another key point was to delegate tasks amongst the team. He said, "If someone can do something to 70% of what you can do, you should hand it off." Last year, with the adaptive bike, I took on many projects, which was overwhelming at times. If I focus on the portal axles, I will need to delegate other tasks, such as CAD model construction, to put in quality work on the portals. The last topic discussed was what the club should be involved with. He noted that the club should work on small projects that constructively contribute to the car. These projects should match their skill and interest level, and they should work on these projects to completion. All these points will influence how we move forward as a project team. We have already scheduled a weekly sprint meeting focused on the integration of subsystems. Also, I have personally dedicated myself more to the portal axles and taken some of my weight off the CAD model. Finally, the club has started working on some small projects they are passionate about, such as merch, social media, the pedal assembly, and the gas tank.

B.4 - Shelby's Interview with Chandra Proffitt

For my stakeholder interview, I had the pleasure of interviewing Chandra Profit, the owner of Profits Resurrection Land Cruiser. She has also been one of the top donors to the SAE Baja project. A question I asked her was, what is the best way to navigate the supply chain? Her response was to order materials and supplies well in advance. She used the steering components the last senior Capstone group ordered in January as an example. It just arrived at her shop earlier this month. Other struggles that she heard about from the previous Capstone group included the limited shop hours. We encountered similar issues in the first couple of months, but the goal is to get the chassis on wheels and move it into the senior design space bay. With this goal, we will initially focus on designing the portals and suspension. She left me with a note to talk to people in the industry who know what they are doing. Also, get your hands on the vehicle as soon as possible. We can do all the research we want, but in the end, someone else has been through this and can save us time. Also, she mentioned not to be afraid of messing something up on the vehicle. Mistakes will happen, and they are learning experiences. The sooner the mistake happens, the more time there is to fix it.

B.5 - Deagen's Interview with Greg Vanderbeek

This stakeholder interview was with Greg Vanderbeek, a professor for the CU partnership program. The focus of this interview was to gain insight into the design of the

Baja car. The first question that was asked was Greg's opinion on the portals. Although a cool idea may benefit the car's performance, Greg suggested that we keep the design simple and keep our end goal (get the car rolling) in mind. A common theme throughout the interview was that we should look at each car component critically and not get caught up in the excessive aspects of the design. Another useful suggestion was to have the group work on multiple subsystems of the car simultaneously to keep the project flow and not get caught up on one aspect of the design. His biggest suggestions were to consider each component critically to find the minimum work to get the car going. A huge resource that would aid in manufacturing would be to utilize personnel with extensive experience working on the car (Kiya Friar, Ben Nienhouse). Greg said that if we could keep things simple and only focus on critical aspects of the design, we would be able to provide a moving vehicle and a solid foundation for future Capstone cohorts.

B.6 - Gable's Interview with Kaiya Firor

The insights gathered from the interview have significant implications for our design and project approach. Firstly, the concern about meeting rules and race requirements underscores the importance of meticulous attention to detail and compliance. Our design process must prioritize a comprehensive understanding of the competition's regulations to ensure we stay within the boundaries. Secondly, communication emerges as a critical issue. To address this, we need to establish robust communication channels within the team, ensuring that even minor changes are effectively conveyed to prevent any disruptions in the design process. The need for more team members dedicated to the project is evident. A larger workforce will enable us to allocate more time to the project, which was a limitation in the previous year. Additionally, an up-to-date CAD model can streamline the design process, accelerating progress. In immediate focus, welding major parts, running the drivetrain, and assembling steering and braking components take precedence. These are integral to the vehicle's functionality, and their timely completion is essential. Prioritizing the mounting brackets for these components is also crucial. Whether portals are worth implementing raises considerations of design challenges and practical advantages. While they may enhance performance and uniqueness, their time-consuming nature must be carefully evaluated. Clearance issues seem unlikely, considering the absence of portals in other teams' designs. Regarding the existing vehicle, it appears that no major aspects require immediate changes. However, temporary pieces should be finalized, and the status of suspension parts, such as control and trailing arms, needs assessment. Ensuring these components are securely in place is essential for the vehicle's reliability and safety.

B.7 - Spencer's Interview with Charlie Schayer

For my stakeholder interview, I interviewed Charlie Schayer. I asked him how the BAJA Capstone team could implement certain materials, design for the competition, and cooperate efficiently and effectively. When asked about what a successful project looked like, he said that success could be measured in many different ways. We are a small start-up program, and we do not have to have a blowout of the water design. Sometimes, a

working design is better than a working design. For example, his team for a phantom wax for skis had many successes but ultimately had a mediocre finished product. This, for us, could look like multiple steps closer to completion or a completed functioning vehicle. He would like to see a completed vehicle from this project. Charlie also said we could integrate carbon fiber into the car's design, mostly in places that don't disrupt or jeopardize the vehicle's integrity. Places such as the steering wheel and body panels were a few that were mentioned. He said that some people in Rady would work with carbon fiber this year and that it could be a part of our vehicle if rules allowed. We do have the tools to work with carbon fiber in Rady. We discussed in class today if we could use our bay in the senior design lab. After the suspension and drivetrain components are finished, we could move the car into that spot, have more room to walk around, etc. This would avoid having a small space in the back of the fabrication shop. The concept of another shop area is still to be determined or approved. As for the steering components, they just arrived at Proffitt's Resurrection Land Cruisers shop. This was roughly a nine-month delivery time. Charlie said to get to the bottom of the cause for this delay. Was it due to the payment not being accepted? Or is it just the vendor being slow or unprofessional? If we could figure out that issue, our wait time to get parts could decrease in our favor. In some BAJA videos, the vehicles roll over the steering wheel. In Charlie's opinion, it could be due to the driver's error, the design of the ramp/course, and the dynamics or specifications of the vehicle. Charlie's solution, if deemed legal, was to create a bull bar or push front bumper that could prevent the motion of rolling head over wheels. He had some input for the BAJA Capstone team about working on a project of this size: to always be in constant communication within the group. Also, after meetings, we could all have a task that we are given to continue the flow of work on the vehicle or in-class work. All in all, great input was given during this interview.

Appendix C – Results from Whole Systems Mapping

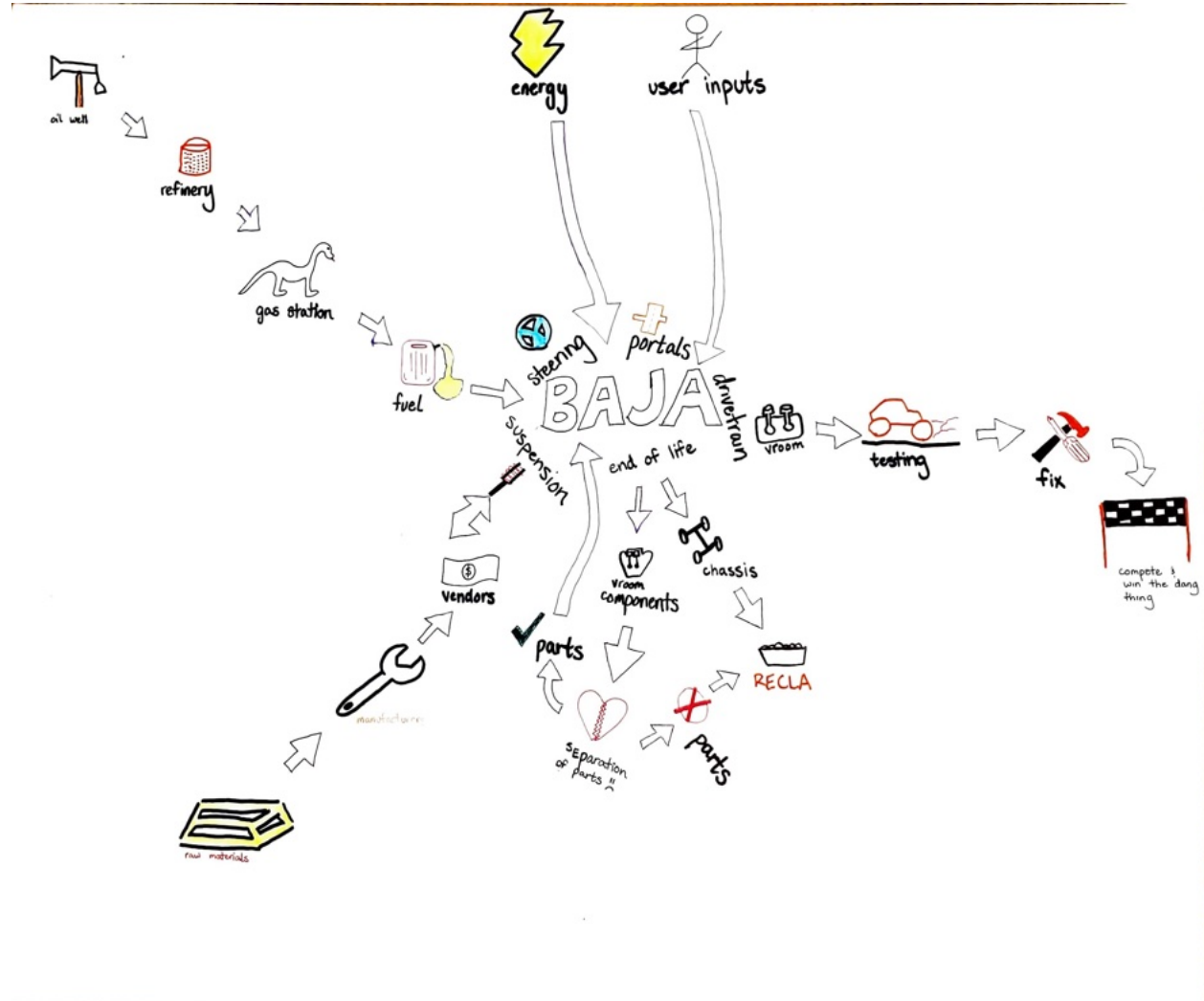


Fig. 4: Whole System Mapping

Appendix D – Design Evaluation Score Sheet Summary

Design Evaluation Score Sheet Summary

Car Number 42

Category	Max Score	Car Rough Score	Car Adjusted Score	Notes to students
Vehicle Level Engineering	35	8		Students will be judged on the creation of design requirements and the ability to meet those requirements, computer aided drafting, analysis, testing and development, manufacturability, serviceability, system integration and how the vehicle works together as a whole. Each of these parts of the engineering product development cycle will be judged within the following subsystems: Suspension, Steering, Brakes, Drivetrain, and Chassis and Ergonomics.
Chassis	20	3		
Ergonomics	10	1		
CVT / Transmission	15	1		
Front drive, prop shaft, front hubs and axles	15	1		
Reduction box, rear hubs and axles	15	1		
Suspension	10	3		
Steering	10	2		
Brakes	10	1		
Innovation (Bonus Points) (Max)	15			
Design Finals (Bonus Points) (Max)	15			
Design Review Briefing (DRB)	10	10		
Total	150	20	30	

CS Scanned with CamScanner
School Name Western Colorado

Design Evaluation Score Sheet

Car Number 42

	Engineering Product Development Cycle	%		Comments
VEHICLE LEVEL ENGINEERING	Vehicle Goals & Cascaded Requirements	10%	2	1st yr. pass tech, more crew performance, need quantitative design, 650 lb weight
	Research/ Development & Component/Subsystem Selection	15%	5	double wishbone front & trailing in rear since most teams do that, look at Pole's
	Full Vehicle Design & CAD	10%	10	good ✓
	Data Collection for Analysis	10%	0	
	Analysis / Calculations / CAE	15%	0	roll-over, side impacts → avg valley for top speeds of other vehicles → not vehicle level
	Testing / Validation	15%	1	run up down hills for an hr
	Correlation of Analysis and Testing	15%	0	
	Assessment Of Vehicle Goals	5%	0	
	Serviceability/ Manufacturability	5%	5	heim joints for quick change removable member to pull engine out
	Score	35	0.8	
BAY LEADER NOTES FOR SUBSYSTEMS	Chassis			
	Ergonomics			
	CVT / Transmission			
	Reduction box, hubs and axles			
	Suspension			
	Steering			
	Brakes			
	Other			
	Total Score	35		

CS Scanned with CamScanner
 School Name Westin Colorado

	Engineering Product Development Cycle	%		Comments
CHASSIS	Requirements	5%	2	- verbal - detail out requirements and show in future
	Research/ Development	5%	1	have some assumptions
	Design / CAD	15%	12	pretty detailed cad
	Data Collection for Analysis	15%	0	should do
	Analysis / Calculations / CAE	25%	14	load inputs should be provided with testing and more analysis to remove conservatism
	Testing / Validation	15%	3	did full vehicle testing
	Correlation of Analysis and Testing	15%	0	
	Serviceability/ Manufacturability	5%	2	2D only bends - automate everything, it has to
	Score	20	3	
ERGONOMICS	Requirements	5%	1	verbal only
	Research/ Development	5%	2	looked at driver sizes measured people
	Design / CAD	15%	7	sketch basic seat, detail out more
	Data Collection for Analysis	15%	0	should do
	Analysis / Calculations / CAE	25%	0	didn't show
	Testing / Validation	15%	3	drove car, got feedback
	Correlation of Analysis and Testing	15%	0	should do
	Serviceability/ Manufacturability	5%	0	should mention
	Score	10	1	
	Total Score	30		

CS Scanned with CamScanner

School Name Western Colorado

Design Evaluation Score Sheet

Car Number 42

	Engineering Product Development Cycle	%		Comments
CVT / TRANSMISSION	Requirements	5%	0	
	Research/ Development	5%	0	
	Design / CAD	15%	15	
	Data Collection for Analysis	15%	0	
	Analysis / Calculations / CAE	25%	0	
	Testing / Validation	15%	0.5	
	Correlation of Analysis and Testing	15%	0	
	Serviceability/ Manufacturability	5%	0.5	
	Score	15	1	
FRONT DRIVE, PROP SHAFT, FRONT HUBS AND AXLES	Requirements	5%	0	
	Research/ Development	5%	0	
	Design / CAD	15%	15	
	Data Collection for Analysis	15%	0	
	Analysis / Calculations / CAE	25%	0	
	Testing / Validation	15%	0.5	
	Correlation of Analysis and Testing	15%	0	
	Serviceability/ Manufacturability	5%	0	
	Score	15	1	
REDUCTION BOX, REAR HUBS AND AXLES	Requirements	5%	0	
	Research/ Development	5%	0	
	Design / CAD	15%	15	
	Data Collection for Analysis	15%	0	
	Analysis / Calculations / CAE	25%	0	
	Testing / Validation	15%	0.5	
	Correlation of Analysis and Testing	15%	0	
	Serviceability/ Manufacturability	5%	0	
	Score	15	1	
	Total Score	45	3	

Goals: - 1st Comp
- Pass tech

CS Scanned with CamScanner
School Name Western Colorado

	Engineering Product Development Cycle	%		Comments
15/13 SUSPENSION	Requirements	5%	2	RC, scrub radius, height note card
	Research/ Development	5%	2	Team survey of competitors
	Design / CAD	15%	12	
	Data Collection for Analysis	15%	0	
	Analysis / Calculations / CAE	25%	10	Racing aspirations software to hit requirements
	Testing / Validation	15%	0	no documentation
	Correlation of Analysis and Testing	15%	0	no documentation
	Serviceability/ Manufacturability	5%	2	talked about how to change
	Score	10	28	
STEERING	Requirements	5%	1	not being limited by steering
	Research/ Development	5%	0	guy who designed not here
	Design / CAD	15%	10	had steering
	Data Collection for Analysis	15%	0	
	Analysis / Calculations / CAE	25%	5	racing aspiration for tiered
	Testing / Validation	15%	0	
	Correlation of Analysis and Testing	15%	0	
	Serviceability/ Manufacturability	5%	2	off the shelf rack
	Score	10	18	
1538 BRAKES	Requirements	5%	2	lockup (rules) part integration
	Research/ Development	5%	1	looked at competitor DRB
	Design / CAD	15%	0	no CAD
	Data Collection for Analysis	15%	0	
	Analysis / Calculations / CAE	25%	0	
	Testing / Validation	15%	3	did drive and lockup, no data
	Correlation of Analysis and Testing	15%	0	no data
	Serviceability/ Manufacturability	5%	2	off the shelf, everything is easy to get to
	Score	10	8	
	Total Score	30		

SP.

- Min people 4 people (full vehicle, chassis, dynamics)
- Talk fast, get through it all
- Follow the process on sheet to get most points
- Binder w/ DRB
 - or powerpoint
 - poster board
- Watch Design Finals of other teams
- Show connection between goals and why certain components were chosen
- Be honest did it work? was the analyze = testing
- Any testing / Data w/ reasoning behind it
 - Even acceleration test with cones and Stopwatch is Good!
- Goals that are quantitative
 - Focus on one section whole presentation should back it up



Scanned with CamScanner